Targeted Local Projections

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Abstract

Local projections (LPs) and structural vector autoregressions (SVARs) are commonly employed to estimate dynamic causal effects of macroeconomic policies. With enough lags as controls, LP estimators have little bias but their variance can increase with the horizon. Because of employing fewer lags or due to local misspecification, SVAR estimators typically incur higher bias, but their variance decreases with the horizon due to exponentiation. We propose to target the LP estimators towards their SVAR counterparts - constructed with fewer lags than LP at each horizon - to reduce their variance at the cost of incurring some bias. The resulting estimator is a linear combination of the LP and SVAR estimators, and we propose choosing it optimally to minimize the mean-squared error of the new estimator. Our simulations show that, under a locally misspecified SVAR model, targeting substantially reduces the LP variance at longer horizons while reducing coverage distortions relative to SVAR when a double bootstrap is employed. In an empirical application, we estimate the impact of a monetary policy shock on macroeconomic aggregates. For many of these aggregates, the corresponding TLP estimates are more precise than their LP counterparts.

Keywords: targeted local projections, SVAR, double bootstrap

JEL Codes: C12, C14, C32

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1 Introduction

Identifying and estimating dynamic causal effects of structural shocks on macroeconomic outcomes is a fundamental goal in macroeconomics. To study these dynamic causal effects, researchers use a variety of tools, including taking to the data theoretical models such as dynamic stochastic general equilibrium models, and other models such as structural vector autoregressions (SVARs) and local projections (LPs).

Local projections were popularized by Jordà (2005)[1] and are widely used to estimate impulse response functions (IRFs) - which coincide with the dynamic causal effects of a structural shock in linear models under appropriate conditioning on lags. A recent overview of the use and inference with LPs is provided in Jordà (2023), Jordà and Taylor (2025), Montiel Olea et al. (2025) and Inoue et al. (2026). LPs are more robust to misspecification than SVARs, as shown in Montiel Olea et al. (2026), and are semi-parametrically efficient under classical assumptions if the number of lags included diverges, as shown in Xu (2026). Therefore, if one is interested exclusively in accurate coverage, either LPs or SVARs with a large number of lags should be employed, as also argued in Montiel Olea et al. (2025).

In this paper, we are interested in pointwise estimation instead. In that case, it is possible to reduce the variance of the LP estimates at the cost of a small bias. This may be useful for applied work because the original LP estimators exhibit variances that increase with the horizon, as suggested by the results in Lusompa (2023), and this is due to the LP regression rather than the true data generating process. The excessive variability of LPs across horizons has prompted researchers to smooth impulse responses across horizons. Plagborg-Møller (2016) proposed smoothing the adjacent changes in LP estimators towards each other, after the LP estimators are obtained, and therefore in two steps. Barnichon and Brownlees (2019) also proposed smooth local projection (SLP) estimators that penalize changes across nearby horizons, but the smoothing is done in one step, via a penalized least-squares objective function. However, smoothing nearby horizons does not guarantee an exponential decay similar to that one would expect if the

[1] An earlier paper by Dufour and Renault (1998) features a similar idea.

true data generating process was a stationary SVAR of finite or infinite order.

In contrast, Ferreira et al. (2023) propose achieving exponential decay by shrinking the LP estimates towards a VAR in a Bayesian framework. To obtain their Bayesian LP (BLP) estimator, they employ a quasi-Bayesian LP approach, where the prior mean of the coefficient is equal to a VAR estimator on a presample, while the prior variance is based on a Minnesota prior, with a shrinkage parameter that maximizes the posterior marginal density of the data. However, due to small samples in macroeconomics, presample VAR estimates are rarely reliable. Additionally, Ferreira et al. (2023) assume that the true data generating process is a finite order VAR, while in reality, it is likely that the model is misspecified - Montiel Olea et al. (2026). To address this issue, González-Casasús and Schorfheide (2025) propose a similar Bayesian procedure, but the shrinkage parameter minimizes the prediction error and their prediction error criterion can also be used to choose among LP and VAR estimators. Their focus is on reduced-form estimation and on multi-step forecasts rather than structural parameters. For point estimation of target structural parameters, it is unclear that minimizing prediction error would be desirable.

Compared to smooth local projections (Plagborg-Møller, 2016; Barnichon and Brownlees, 2019), TLP differs in both its regularization target and its treatment of horizons. SLP penalizes differences across adjacent horizons, shrinking the IRFs toward each other, without a data-driven target, while TLP shrinks LPs toward the SVAR IRFs. SLP also applies a single smoothing parameter across all horizons and creates distortions at boundary horizons; TLP allows shrinkage to vary by horizon and does not directly impose cross-horizon smoothness. Compared to BLP (Ferreira et al., 2023), TLP uses the full sample for both LP and SVAR estimation, whereas BLP relies on a presample VAR as the prior. Additionally, BLP assumes correct VAR specification and our simulations show that under misspecification, it exhibits severe coverage distortions at shorter horizons. Finally, TLP is a linear combination of LP and SVAR with transparent, data-driven weights, while BLP is not a linear combination of the two.

There are a few other papers that also consider taking linear combinations of LP and SVAR. Ho et al. (2024) consider prediction pools where multiple estimators are averaged to minimize the weighted average log score function for forecasts, conditional on the structural shock of interest. Hounyo and Jung (2026) propose to first take horizon-specific linear combinations of multiple LP and SVAR estimators separately, then take a linear combination of these two, with weights that minimize prediction error or R^2. The motivation for TLP is different from that of Ho et al. (2024) and Hounyo and Jung (2026): we want to reduce the variance of the LP estimators, at the cost of incurring a small bias, and therefore the risk function we minimize is the mean-squared error of TLP. In another related but distinct contribution, Chen et al. (2026) also propose a linear combination of LP and SVAR estimators to minimize the risk of the resulting estimator - variance or mean-squared error; however, their asymptotic theory is in the context in which both LP and SVAR do not exhibit asymptotic bias. They start from a general data generating process that can be approximated with sufficiently many lags that diverge with the sample size at the appropriate rate, and employ a sieve AR bootstrap with bias correction to estimate their proposed horizon-specific weights. Compared to our estimator, Chen et al. (2026) focus on finite sample rather than asymptotic trade-offs; therefore, their approach complements ours.

In the case of local misspecification, the SVAR estimates will exhibit asymptotic bias, distorting coverage of TLP as well. As discussed in Montiel Olea et al. (2025), impossibility results from microeconometrics - Pratt (1961), Armstrong and Kolesár (2018) - suggest that coverage distortions may be unavoidable.[2] However, the coverage distortions of TLP will be smaller than those of the SVAR because the bias of TLP is smaller.

For inference, we propose a Mean Symmetric Double Bootstrap (MSDB) procedure that provides near-nominal coverage for LP, SLP, and TLP when the SVAR misspecification is sufficiently small, as shown in our simulations. The procedure uses a nested bootstrap structure: the first level generates bootstrap samples via moving-block resampling of SVAR residuals, while the second level provides variance estimates needed for studentization, which are necessary because the delta method performs poorly for SVAR

[2] Armstrong and Kolesár (2018) provide a general procedure for adaptive estimators to balance efficiency and robustness, and suggest reporting both best- and worst-case coverage, but do not consider time series set-ups.

and TLP impulse responses in finite samples. Two innovations are central to minimizing coverage distortions for TLP and VAR. First, we center the bootstrap t-statistics around the bootstrap mean rather than the SVAR-implied impulse response to remove further finite sample distortions. Second, we construct symmetric confidence intervals (Hall, 1988), which help to correct finite sample asymmetry in the VAR and TLP distributions.

It is worth noting that this bootstrap replicates the variance of the SVAR and TLP estimates, but not the asymptotic bias when the SVAR is locally misspecified. Intuitively, the first level bootstrap is centered at the SVAR estimates, so the difference between the bootstrap estimates and the SVAR estimates is centered around zero, while the difference between SVAR and TLP estimates and the true IRFs are centered at their respective asymptotic bias. Therefore, the MSDB studentized confidence intervals will still be shifted due to bias; however, the shift is smaller for TLP because its asymptotic bias is smaller. In simulations featuring recursively identified SVARs with local misspecification, we demonstrate that TLP substantially reduces variance relative to LP and mean-squared error at longer horizons relative to both LP and SVAR, while maintaining superior coverage to that of the SVAR, and near-nominal coverage provided the misspecification is not too large.

We revisit the empirical application in Ferreira et al. (2023) and estimate the IRFs of a monetary shock on real GDP, real consumption, real investment, total hours worked, real wages and the GDP deflator. For several but not all of these variables, the LP and SVAR estimates are markedly different at higher horizons, suggesting that the SVAR may be misspecified, as also noted in Ferreira et al. (2023). In those cases, TLP estimates still have smaller variance than LP, even though more weight is placed on LP. When the LP and SVAR estimates are close to each other, such as for IRFs of a monetary policy shock on hours worked, more weight is placed on SVAR and the TLP variance reductions relative to LP are larger.

The remainder of this paper proceeds as follows. Section 2 presents the baseline estimators and introduces the targeted local projections estimator. Section 3 derives the optimal shrinkage parameter by minimizing mean-squared error. Section 4 describes the bootstrap procedure. Section 5 evaluates the finite-sample performance of TLP relative to existing methods. Section 6 contains the empirical application. Section 7 concludes. Appendix A discusses the importance of projecting out controls before targeting. Appendix B contains proofs, Appendix C further simulations, and Appendix D further empirical results.

2 Model set-up

2.1 Data Generating Process

We consider the data generating process to be a local-to-VAR model[3] that is locally misspecified as in Montiel Olea et al. (2026):

$$y_t = Ay_{t-1} + \Gamma\left(I + T^{-\zeta}\alpha(L)\right)\epsilon_t, \tag{1}$$

where $y_t = (y_{1,t}, \ldots, y_{k,t})'$ is the data, the structural shocks are $\epsilon_t = (\epsilon_{1,t}, \ldots, \epsilon_{k,t})'$, $\alpha(L) = \sum_{\ell=1}^{\infty} \alpha_\ell L^\ell$ is a $k \times k$ lag polynomial, and T denotes the sample size. The parameter of interest is the impulse response at horizon h of the variable $y_{i,t+h}$ to the shock $\epsilon_{j,t}$ for some indices $i, j \in \{1, \ldots, k\}$. The assumption below is adapted from Montiel Olea et al. (2026), setting for simplicity the number of shocks equal to the number of variables.[4]

Assumption 1. *For each T, $\{y_t\}_{t\in\mathbb{Z}}$ is the stationary solution to equation (1), given the following restrictions on parameters and shocks:*

i) ϵ_t is a strictly stationary martingale difference sequence with respect to the filtration $\mathcal{F}_{t-1} = \sigma\{\epsilon_{t-1}, \epsilon_{t-2}, \ldots\}$; $\mathrm{Var}(\epsilon_t) = D = \mathrm{diag}(\sigma_1^2, \ldots \sigma_k^2)$; $\sigma_j^2 > 0$ for all $j = 1, \ldots, k$; $\mathbb{E}\|\epsilon_t\|^d < \infty$ for $d = 4 + 2\delta$, for some $\delta > 0$; $\sum_{\ell,\tau=1}^{\infty} \|\mathrm{Cov}(\epsilon_t \otimes \epsilon_t, \epsilon_{t-\ell} \otimes \epsilon_{t-\tau})\| < \infty$.

ii) All eigenvalues of A are inside the unit circle and different than zero.

iii) Γ is a $k \times k$ lower triangular matrix with 1's on the diagonal.

[3] We refer to the SVAR as VAR from now onward.

[4] This can be generalized to having more or fewer shocks as long as the shocks of interest are identified via appropriate ordering.

iv) $S = \mathrm{Var}(\tilde{y}_t)$ *is non-singular, where* $\tilde{y}_t = (I - AL)^{-1}\Gamma\epsilon_t$ *is the stationary solution to (1) when* $\alpha(L) = 0$.

v) $\alpha(L)$ *is absolutely summable.*

vi) $\zeta > 1/4$.

This assumption ensures that the process $\tilde{y}_t$ around which the VAR process y_t is locally misspecified is stationary. Assumption 1(iii) ensures that the dynamic causal effects of the shocks $\epsilon_{i,t}$ are identified through ordering, a common assumption for identification.

The impulse response at horizon $h = \{0, \ldots, H\}$ is defined as

$$\begin{aligned} \beta_h &= E\left[y_{i,t+h} \mid \epsilon_{j,t} = 1\right] - E\left[y_{i,t+h} \mid \epsilon_{j,t} = 0\right] \\ &= J_i'\left(A^h\Gamma + T^{-\zeta}\sum_{\ell=1}^{h} A^{h-\ell}\Gamma\alpha_\ell\right)J_j, \end{aligned} \tag{2}$$

where J_i and J_j are the selection vectors of dimensions $k \times 1$ with 1 in the position of i or j and 0 otherwise. Note that the IRF we consider at each horizon is a scalar. If the true data generating process (DGP) has more than one lag $q > 1$, we can transform it into a local-to-VAR(1) using the companion matrix, and therefore consider without loss of generality the local-to-VAR(1) data generating process. We also omit in the DGP above the intercept for simplicity; however, this can be included without further complications besides additional notation.

2.2 Baseline Estimators

Local projections (LP). The LP estimator is the coefficient $\hat{\beta}_h^{LP}$ in the regression

$$y_{i,t+h} = \beta_h y_{j,t} + \sum_{s=1}^{k}\sum_{\tau=1}^{p}\gamma_{s,t-\tau}y_{s,t-\tau} + u_{i,t+h}^h. \tag{3}$$

where $u^h_{i,t+h}$ is the LP error, and p is the number of lags included as controls. Stacking $y_{i,t+h}$, $y_{j,t}$, controls and $u^h_{i,t+h}$ over time in the vectors $Y_{i,t+h}$, $Y_{j,t}$, W and $U^h_{i,t+h}$ yields

$$Y_{i,h} = Y_j \beta_h + W\delta_h + U_i^h, \tag{4}$$

where W is a $(T-h) \times (kp)$ matrix collecting the lagged controls, and δ_h stacks $\gamma_{s,t-\tau}$ appropriately over s, τ. Note that the LP estimator we consider here is lag-augmented with $p \geq 1$ lags.

Projecting out controls from both $Y_{i,h}$ and Y_j:

$$\underbrace{M_W Y_{i,h}}_{Y_h} = \underbrace{M_W Y_j}_{X} \beta_h + \underbrace{M_W U_i^h}_{e_h}, \tag{5}$$

where $M_W = I - W(W'W)^{-1}W'$ yields the simplified model:

$$Y_h = X\beta_h + e_h. \tag{6}$$

Projecting out controls is not strictly necessary for LP estimation, but we do so because the coefficients δ_h are typically nuisance parameters, and we only care about the dynamic causal effects. Regularizing the latter without penalizing the nuisance parameters induces further bias in the coefficients on the control variables without any clear benefit.[5]

Structural vector autoregression (VAR). The SVAR estimator $\hat{\beta}_h^{VAR}$ is the response of $y_{i,t+h}$ to the j-th innovation, obtained via Cholesky decomposition:

$$\hat{\beta}_h^{VAR} = J_i' \hat{A}^h \hat{\Gamma} J_j, \tag{7}$$

where $\hat{A}$ is the VAR coefficient matrix estimate obtained via OLS equation by equation. $\hat{A} = \left(\sum_{t=1}^T y_t y_{t-1}'\right)\left(\sum_{t=1}^T y_{t-1} y_{t-1}'\right)^{-1}$, and $\hat{\Gamma}$ is the lower triangular Cholesky factor of the residual covariance matrix $\hat{\Sigma}^{VAR} = \frac{1}{T}\sum_{t=1}^T \hat{u}_t \hat{u}_t'$, with $\hat{u}_t = y_t - \hat{A} y_{t-1}$.

[5] Appendix A clarifies the source of this additional bias.

Asymptotic properties. Under the local-to-VAR framework with $\zeta > 1/4$, Montiel Olea et al. (2026) show that LP is doubly robust in the sense that it has no asymptotic bias. In contrast, VAR suffers from asymptotic bias when $\zeta \in (1/4, 1/2]$. This is stated in their Propositions 3.1 and 3.2, which assume that the number of lags is the same in the LP and VAR and equal to that in the true DGP. However, the result generalizes to the case in which LP contains more lags than the true number of lags, and this is stated in Theorem 1 below.

Theorem 1. *Under Assumption 1,* $\mathbb{E}(\hat{\beta}_h^{LP} - \beta_h) = o(T^{-1/2})$, *while* $\mathbb{E}(\hat{\beta}_h^{VAR} - \beta_h) = T^{-\zeta}\, aBias_h + o(T^{-1/2} + T^{-\zeta})$, *where*

$$aBias_h = tr\left(S^{-1}\Psi_h \Gamma \sum_{\ell=1}^{\infty} \alpha_\ell D \Gamma'(A')^{\ell-1} \right) - J_i' \sum_{\ell=1}^{h} A^{h-\ell}\Gamma \alpha_\ell J_j,$$
$$\Psi_h = \sum_{\ell=1}^{h} A^{h-\ell}\Gamma_j J_i' A^{\ell-1}.$$

When $1/4 < \zeta \leq 1/2$, $\mathrm{E}(\hat{\beta}_h^{VAR}) - \beta_h = O(T^{-\zeta})$, and the VAR estimator has an asymptotic bias, while when $\zeta > 1/2$, the VAR does not have an asymptotic bias.

Next, we impose mixing conditions on the structural errors.

Assumption 2. *Assume that ϵ_t is an α-mixing process of size $-r/(r-2)$, for $r = 2 + \delta$ and the same δ as in Assumption 1.*

The mixing condition above is used to verify the WLLN and CLT for near-epoch dependent processes (also mixingales) and to show that in this case, the asymptotic LP variance does not depend to first order on the misspecification.[6]

Theorem 2. *Under Assumptions 1-2, the LP and SVAR estimators are jointly asymptotically normal:*

$$\sqrt{T}\begin{pmatrix} \hat{\beta}_h^{LP} - \beta_h \\ \hat{\beta}_h^{VAR} - \beta_h \end{pmatrix} \xrightarrow{d} \mathcal{N}\left(\begin{pmatrix} 0 \\ \text{aBias}_h \end{pmatrix}, \begin{pmatrix} \Sigma_h^{LP} & \Sigma_h^{COV} \\ \Sigma_h^{COV} & \Sigma_h^{VAR} \end{pmatrix} \right), \tag{8}$$

[6] Note that, given the misspecification, the martingale difference assumption on the errors is insufficient to verify the CLT, and, as Xu (2026) notes, more moment bounds are needed to ensure the asymptotic variances exist, and imposing those would make the assumptions less transparent.

where Σ_h^{LP} and Σ_h^{VAR} are the limiting asymptotic variances of LP and VAR described in (40) *and* (41) *in Appendix B, and where Σ_h^{COV} denotes the asymptotic covariance between LP and VAR.*

Note that Σ_h^{LP}, Σ_h^{COV} and Σ_h^{VAR} are given in Corollary A.2 of Montiel Olea et al. (2026) for i.i.d. structural errors, and they show that when the number of lags used in LP and VAR is the same, then $\Sigma_h^{VAR} = \Sigma_h^{COV}$. However, in this paper we allow for more lags in LP than in VAR, and conditional heteroskedasticity, in which case their exact form depends on the data generating process. Nevertheless, they do not depend on the misspecification when $\zeta = 1/2$, and will be estimated by the bootstrap.

Bias - variance trade-off. The asymptotic properties of LP and VAR suggest a natural trade-off. In terms of bias, LP is consistent under local misspecification, while VAR exhibits an asymptotic bias of order $T^{-\zeta}$. In terms of variance, the LP variance can increase with the horizon due to the accumulated shocks in $u_{i,t+h}^{h}$ obtained by backward substituting the VAR model in (1). On the other hand, under no misspecification and with i.i.d. normal structural shocks, the VAR estimator is the maximum likelihood estimator and is therefore more efficient than LP. Theorem 2 shows that the local misspecification shifts the VAR distribution but the asymptotic variance stays the same. Under conditional heteroskedasticity, it is not known how the VAR and LP variances compare, but it is likely that VAR still has lower variance at higher horizons, as this variance eventually declines under stationarity because $A^h \to 0$ as $h \to \infty$, while this is not the case for LP.

These observations motivate targeting LP towards the VAR: the variance of LP can be reduced by shrinking toward VAR, accepting a small amount of bias in exchange for lower variance. Additionally, Ludwig (2026) shows that LPs with p lags can be represented as combinations of equal and higher order VARs up to order $p + h - 1$, and therefore that, for a given number of lags, the LPs use a more complex model compared to VARs of equal order p or lower. This further motivates targeting LP toward the VAR.

2.3 Targeted Local Projections

We propose a targeted local projections (TLP) estimator that shrinks LP impulse responses toward their VAR counterparts at each horizon. This subsection defines the estimator, derives its closed-form solution, and discusses its relationship to existing methods.

Estimator. The TLP estimator $\hat{\beta}_h^{\text{TLP}}$ minimizes a penalized LP objective function at each horizon over $\tilde{\beta}_h$:

$$Q_n(\tilde{\beta}_h) = \big(Y_h - X\tilde{\beta}_h\big)^2 + \lambda_h\big(\tilde{\beta}_h - \hat{\beta}_h^{VAR}\big)^2, \tag{9}$$

where $\lambda_h > 0$ controls the degree of shrinkage. Solving for $\tilde{\beta}_h$ yields:

$$\hat{\beta}_h^{\text{TLP}} = (X'X + \lambda_h)^{-1}(X'Y_h + \lambda_h\hat{\beta}_h^{VAR}), \tag{10}$$

where $X'X$ is a scalar after projecting out controls. This expression can be rewritten as a linear combination of LP and VAR estimators:

$$\begin{aligned}
\hat{\beta}_h^{\text{TLP}} &= (X'X + \lambda_h)^{-1}X'Y_h + (X'X + \lambda_h)^{-1}\lambda_h\hat{\beta}_h^{VAR} \\
&= (X'X + \lambda_h)^{-1}X'X\,\hat{\beta}_h^{LP} + \big(1 - (X'X + \lambda_h)^{-1}X'X\big)\hat{\beta}_h^{VAR} \\
&= v(\lambda_h)\hat{\beta}_h^{LP} + (1 - v(\lambda_h))\hat{\beta}_h^{VAR},
\end{aligned} \tag{11}$$

where $v(\lambda_h) = (X'X + \lambda_h)^{-1}X'X$ is a scalar weight and is by construction in the interval $(0, 1)$. The TLP impulse response is therefore a linear combination of LP and VAR. Note that when $\lambda_h \to 0$, the weight $v(\lambda_h) \to 1$ and TLP reduces to LP, while when $\lambda_h \to \infty$, the weight $v(\lambda_h) \to 0$ and TLP reduces to VAR.

Horizon-specific weights. Unlike model averaging approaches that apply uniform weights across horizons (Montiel Olea et al., 2026), TLP allows the shrinkage parameter λ_h to vary with the horizon. This flexibility is desirable because the relative performance of LP and VAR differs across horizons. At short horizons, LP estimates have low bias and are therefore more informative. At higher horizons, LP variance increases due to accumulated shocks in the regression error, making it preferable to place greater weight on VAR. Moreover, Ludwig (2026) shows that the complexity of LPs relative to VARs of equal or lower order increases with the horizon, further motivating horizon-specific shrinkage.

2.4 Comparison to Other Estimators

Bayesian local projections (BLP). In the case of no additional controls and for a given λ_h, the mean of the TLP estimator is equal to the BLP posterior mean in Ferreira et al. (2023). Despite this connection, three distinctions are worth noting: BLP assumes the VAR is correctly specified; it does not project out controls, so the BLP impulse responses are not necessarily linear combinations of LP and VAR impulse responses; it relies on a presample VAR, while TLP uses the full sample.

Smooth local projections (SLP). TLP and SLP (Barnichon and Brownlees, 2019; Plagborg-Møller, 2016) both employ ridge-type regularization to reduce the variance of impulse response estimates. However, they differ in their regularization targets and in how shrinkage operates across horizons. This subsection describes the SLP estimator and highlights these distinctions. As with TLP, we project out control variables before estimation to avoid targeting nuisance parameters.

Stack the projected regressor X and outcome Y_h across horizons $h = 0, \ldots, H$:

$$\tilde{X} = I_H \otimes X, \qquad \tilde{Y} = \begin{pmatrix} Y_0 \\ \vdots \\ Y_H \end{pmatrix}, \qquad \tilde{\epsilon} = \begin{pmatrix} \epsilon_0 \\ \vdots \\ \epsilon_H \end{pmatrix}, \tag{12}$$

so that $\tilde{Y} = \tilde{X}\beta + \tilde{\epsilon}$, where $\beta = (\beta_0, \ldots, \beta_H)'$ collects the $H+1$ impulse responses across all horizons. The simplest version of SLP penalizes second differences of adjacent impulse responses. Define the second-order difference matrix:

$$L = \begin{bmatrix} 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & -2 & 1 \end{bmatrix}_{(H+1-2)\times(H+1)} . \tag{13}$$

The SLP estimator minimizes the penalized objective:

$$Q_n(\beta) = \|\tilde{Y} - \tilde{X}\beta\|^2 + \tilde{\lambda}\|L\beta\|^2, \tag{14}$$

where $\tilde{\lambda} \geq 0$ controls the degree of smoothing. Setting $P = L'L$, we obtain:

$$\hat{\beta}^{\text{SLP}} = (\tilde{X}'\tilde{X} + \tilde{\lambda}P)^{-1}\tilde{X}'\tilde{Y}. \tag{15}$$

In the limit $\tilde{\lambda} \to \infty$, the penalty forces $L\beta \to 0$, which shrinks the impulse response function toward a straight line. Conversely, when $\tilde{\lambda} = 0$, SLP reduces to LP. For intermediate values, SLP smooths adjacent horizons toward each other.

Selection of the SLP smoothing parameter. Two approaches have been proposed for selecting $\tilde{\lambda}$:

(i) *Cross-validation.* Barnichon and Brownlees (2019) recommend k-fold cross-validation based on Racine (1997). In their empirical application, they use leave-one-out cross-validation with the standard shortcut formula that avoids refitting for each held-out observation.

(ii) *Unbiased risk estimator.* Plagborg-Møller (2016) proposes minimizing an unbiased risk estimator (URE):

$$\min_{\tilde{\lambda}} \hat{R}(\tilde{\lambda}) = T\|\hat{\beta}^{\text{SLP}} - \hat{\beta}^{LP}\|_W^2 + 2\,\mathrm{tr}\{W\psi(\tilde{\lambda})\hat{\Sigma}\}, \tag{16}$$

where $\hat{\beta}^{LP} = (\hat{\beta}_0^{LP}, \ldots, \hat{\beta}_H^{LP})'$, $||a||_W^2 = a'Wa$, W is a weighting matrix (typically

$W = I_H$), $\psi(\tilde{\lambda}) = (\tilde{X}'\tilde{X} + \tilde{\lambda}P)^{-1}\tilde{X}'\tilde{X}$, and $\hat{\Sigma}$ is a HAC estimator of the long-run covariance matrix of LP that accounts for serial correlation across horizons.

In unreported simulations, we find that both criteria yield similar results. For comparability with TLP, we select $\tilde{\lambda}$ by minimizing the risk criterion (16) throughout. This criterion is analogous to the risk function used for TLP and discussed in section 3.

Key differences between SLP and TLP. Three features distinguish TLP from SLP. First, SLP smooths across horizons, while TLP shrinks toward the VAR-implied impulse responses. Second, by construction, SLP applies a single smoothing parameter $\tilde{\lambda}$ uniformly across all horizons, while TLP allows the shrinkage parameter λ_h to vary with the horizon, reflecting the different bias - variance trade-offs at short and long horizons. As the variance of LP can increase with the horizon, while the variance of VAR decreases with the horizon and the asymptotic bias of VAR remains the same across horizons, horizon-specific weights are desirable. Third, SLP smooths adjacent horizons toward each other, which creates distortions at boundary horizons.

3 Selecting the Shrinkage Parameter

The TLP estimator depends on the shrinkage parameter λ_h, which governs the weight placed on LP versus VAR at each horizon. A key advantage of TLP is that the optimal weight we propose here has a closed-form solution and can be computed directly from the data. This section derives this solution by minimizing the mean-squared error of TLP and is related to the focused information criterion of Claeskens and Hjort (2003).

Objective function. We select the shrinkage parameter at each horizon by minimizing an asymptotically unbiased estimator of the mean-squared error:

$$R_h(\lambda_h) = T\,\mathbb{E}\big[\big(\hat{\beta}_h^{\mathrm{TLP}} - \beta_h\big)^2\big], \tag{17}$$

where recall that β_h is the true impulse response. The risk decomposes into squared bias and variance:

$$R_h(\lambda_h) = \underbrace{T\big(\mathbb{E}[\hat{\beta}_h^{\text{TLP}}(\lambda_h) - \beta_h]\big)^2}_{\text{bias}^2(\lambda_h)} + \underbrace{T\,\mathbb{E}\big[\big(\hat{\beta}_h^{\text{TLP}}(\lambda_h) - \mathbb{E}[\hat{\beta}_h^{\text{TLP}}(\lambda_h)]\big)^2\big]}_{Var(\lambda_h)}. \tag{18}$$

Assume we have consistent estimators $\hat{\Sigma}_h^{\mu} \xrightarrow{p} \Sigma_h^{\mu}$, for $\mu \in \{LP, VAR, COV\}$. Consistent estimators for LP can be obtained by standard HAC estimators; however, we use the bootstrap. Consistent estimators for the rest of the quantities are discussed in the next subsection.

Theorem 3. *Asymptotically unbiased risk criterion. Under Assumptions 1-2, for* $\zeta = 1/2$*, the feasible risk criterion:*

$$\hat{R}_h(\lambda_h) = (1 - v(\lambda_h))^2 T(\hat{\beta}_h^{LP} - \hat{\beta}_h^{VAR})^2 + (2v(\lambda_h) - 1)\hat{\Sigma}_h^{LP} + 2(1 - v(\lambda_h))\hat{\Sigma}_h^{COV}. \tag{19}$$

is asymptotically unbiased: $\sup_{\lambda_h \in (0,\infty)} |\mathbb{E}[\hat{R}_h(\lambda_h)] - R_h(\lambda_h)| \to 0.$

Theorem 3 derives an asymptotically unbiased risk criterion under local misspecification of order $T^{-1/2}$, the only one under which a meaningful asymptotic bias - variance trade-off is possible. The criterion (19) is quadratic in $v(\lambda_h)$, so minimization yields a unique closed-form solution.

Let the optimal shrinkage parameter minimize the feasible risk criterion:

$$\hat{\lambda}_h = \arg \min_{\lambda_h \in (0,\infty)} \hat{R}_h(\lambda_h),$$

when the implied weights $v(\hat{\lambda}_h)$ are positive, and $\hat{\lambda}_h = \infty$, so $v(\hat{\lambda}_h) = 0$, otherwise. We then obtain the following closed-form expression for the TLP weights:

Theorem 4. *Optimal weight. Under Assumptions 1-2, for* $\zeta = 1/2$*,*

$$v(\hat{\lambda}_h) = max\left[1 - \frac{\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{COV}}{T(\hat{\beta}_h^{LP} - \hat{\beta}_h^{VAR})^2}, 0\right]. \tag{20}$$

If, furthermore, the LP and VAR have the same number of lags, and the structural shocks *are i.i.d., then*

$$v(\hat{\lambda}_h) = max\left[1 - \frac{\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{VAR}}{T(\hat{\beta}_h^{LP} - \hat{\beta}_h^{VAR})^2}, 0\right]. \tag{21}$$

The result (21) follows from Corollary A.2 of Montiel Olea et al. (2026), which shows that with the same number of lags and i.i.d. structural shocks, $\Sigma_h^{VAR} = \Sigma_h^{COV}$. It is worth noting:

(i) If $1/4 < \zeta < 1/2$, the bias term dominates the variance terms, the weight $v(\hat{\lambda}_h)$ approaches one, and TLP places all weight on LP, avoiding the biased VAR.

(ii) If $\zeta > 1/2$, the VAR has no asymptotic bias, the limit expectation of $T(\hat{\beta}_h^{LP} - \hat{\beta}_h^{VAR})^2$ is $\hat{\Sigma}_h^{LP} + \hat{\Sigma}_h^{VAR} - 2\hat{\Sigma}_h^{COV}$, which is equal to the numerator in (21) $\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{COV}$ if $\hat{\Sigma}_h^{VAR} = \hat{\Sigma}_h^{COV}$. However, even in the latter case, because the weights are random, not all weight will be placed on VAR since the maximum is with positive probability not achieved at 0.

(iii) If $0 < \zeta \leq 1/4$, a case not covered by our assumptions, both LPs and VAR have asymptotic bias, of a larger order than the variance terms, and TLP asymptotically places all weight on LP.

Comparison to uniform model averaging. An alternative to minimizing risk is to choose weights based on theoretical misspecification bounds. Montiel Olea et al. (2026) propose selecting the weight based on $M = \|\alpha(L)\|$, the norm of the misspecification polynomial, which applies to all horizons uniformly. Their Corollary 4.2 states that the optimal weight ω between LP and VAR should be:

$$\omega = \frac{M^2}{M^2 + 1}. \tag{22}$$

Comparing equations (20) and (22), it can be noted that for large misspecifications $0 < \zeta < 1/2$, TLP asymptotically places all weight on LP.

Comparison to Chen et al. (2026). In their asymptotic theory, Chen et al. (2026) consider averaging LP and VAR when they both do not exhibit asymptotic bias. Because Chen et al. (2026) do not shrink LP towards the VAR, but seek the linear combination of LP and VAR that minimizes the mean-squared error of their estimator, their formula is different from ours under a correctly specified VAR.[7]

4 Variance estimation

Constructing confidence intervals for impulse response functions requires reliable variance estimates. For LP, standard variance estimators are available. For VAR impulse responses, however, it is known that the delta method performs poorly in finite samples. TLP inherits these challenges since it combines LP with VAR estimates. We therefore employ the bootstrap.

We propose an MSDB procedure for inference on LP, VAR, SLP, and TLP estimators. The double bootstrap is employed to calculate *studentized* confidence intervals, which are important to achieve asymptotic refinements when the misspecification is not too large (when $\zeta > 1/2$).[8] As discussed earlier, for $\zeta = 1/2$, the VAR and TLP estimators suffer from an asymptotic bias. The residual bootstrap we employ does not replicate this bias as it is centered around the VAR and TLP estimators, but it does replicate their variance. This means that the bootstrap t-statistics used for studentization are centered around zero, and the confidence intervals are shifted due to asymptotic bias, distorting coverage. The benefit of the bootstrap in this case is that by centering around zero, these confidence intervals are not further shifted.

Our procedure shares similarities with the VAR-based resampling scheme in Montiel Olea and Plagborg-Møller (2021), Montiel Olea et al. (2026) and Chen et al. (2026), but differs in several aspects:

[7]Note that while their theory is derived under correct specification, their estimator can also be applied to misspecified cases, and their simulations are under less severe misspecifications than ours.

[8]Recently, Cavaliere et al. (2024) have demonstrated that in the presence of asymptotic bias, one can use either pre-pivoting or double bootstrap to restore coverage. Their setting does not apply to our bootstrap because neither the single nor the double bootstrap we employ can replicate the asymptotic bias.

(i) We use the recursive residual moving block bootstrap (Brüggemann et al., 2016), while Montiel Olea et al. (2026) use a recursive residual i.i.d. bootstrap, and Chen et al. (2026) propose a residual i.i.d. bootstrap after employing a sieve autoregression. All of these are valid under no misspecification.

(ii) We employ a double bootstrap rather than a single bootstrap. A single bootstrap does not provide reliable variance estimates needed for studentization, since the delta method performs poorly in small samples. The double bootstrap provides consistent variance estimates for standardization; intuitively, the second layer of bootstrap is generated with the true DGP being the first level bootstrap, so it has no bias relative to that DGP, rendering consistent estimates of the VAR variances and the covariance between LP and VAR.

(iii) When constructing t-statistics, we subtract the bootstrap mean rather than the VAR-implied impulse response. This mean-centering removes finite sample bias in the numerator of the bootstrap t-statistics.

(iv) We use symmetrized confidence intervals (Hall, 1988). Asymmetric intervals work well for LP, but they do not account for the skewness in the VAR and TLP t-statistics.

First level bootstrap. Brüggemann et al. (2016) showed that if Assumptions similar to 1-2 hold with $\alpha(L) = 0$, so the model is correctly specified, the recursive moving block bootstrap is valid under conditional heteroskedasticity of unknown form for the VAR estimates. This bootstrap is the one we employ and it proceeds in the following steps: (i) estimate the reduced-form VAR and obtain the residuals $\hat{u}_t$; (ii) divide the residual series into overlapping blocks of length $\ell = T^{1/3}$, following Gonçalves and Kilian (2004); (iii) draw blocks with replacement and paste them together until the series $\{u_t^{(b)}\}_{t=1}^{T}$ reaches length T; (iv) reconstruct the bootstrap sample recursively $y_t^{(b)} = \hat{A} y_{t-1}^{(b)} + u_t^{(b)}$; (v) in each bootstrap replication, estimate impulse responses using both VAR and LP.

Second level bootstrap The double bootstrap proceeds as follows:

i) Treat the original sample as $b_1 = 1$ and generate $B_1 - 1$ bootstrap samples using the moving-block procedure described above. For each sample $b_1 = 1, \dots, B_1$, estimate impulse responses $\hat{\beta}^{\mu}_{h,b_1}$ at each horizon h for $\mu \in \{LP, VAR, SLP\}$.

ii) For each first-level bootstrap sample, apply the same moving-block resampling procedure to generate B_2 second-level bootstrap replications, now treating the first-level sample as the data. This yields $\hat{\beta}^{\mu}_{h,b_1,b_2}$ for $b_2 = 1, \dots, B_2$.

iii) Compute the variances and the covariance between LP and VAR for $b_1 > 1$ from second-level replications:

$$\hat{\Sigma}^{\mu}_{h,b_1} = \widehat{\mathrm{Var}}_{b_2}\left(\hat{\beta}^{\mu}_{h,b_1,b_2}\right), \qquad \hat{\Sigma}^{COV}_{h,b_1} = \widehat{\mathrm{Cov}}_{b_2}\left(\hat{\beta}^{LP}_{h,b_1,b_2}, \hat{\beta}^{VAR}_{h,b_1,b_2}\right). \tag{23}$$

For LP and VAR, these quantities are sufficient to construct the t-statistics. TLP is constructed from LP and VAR using the weights described below. SLP requires additional steps to obtain variance estimates.

TLP. To compute the weights $v(\hat{\lambda}_h)$, we need consistent estimates of $\hat{\Sigma}^{LP}_h$, $\hat{\Sigma}^{VAR}_h$, and $\hat{\Sigma}^{COV}_h$. We obtain these by averaging the double-bootstrap estimates across first-level bootstrap samples:

$$\hat{\Sigma}^{\mu}_{h} = \frac{1}{B_1}\sum_{b_1=1}^{B_1}\hat{\Sigma}^{\mu}_{h,b_1}, \quad \mu \in \{LP, VAR\}, \qquad \hat{\Sigma}^{COV}_{h} = \frac{1}{B_1}\sum_{b_1=1}^{B_1}\hat{\Sigma}^{COV}_{h,b_1}. \tag{24}$$

These are used to compute the weights $v(\hat{\lambda}_{h,b_1})$ using equation (20), with first level bootstrap used for LP and VAR estimates. We re-estimate the bias components of the weights of TLP in each first level bootstrap replication. The bootstrap replications of TLP are then

$$\hat{\beta}^{TLP}_{h,b_1}(\hat{\lambda}_{h,b_1}) = v(\hat{\lambda}_{h,b_1})\hat{\beta}^{LP}_{h,b_1} + (1 - v(\hat{\lambda}_{h,b_1}))\hat{\beta}^{VAR}_{h,b_1}, \tag{25}$$

and the variance of each first-level replication is computed as the average variance of the TLP estimates across the second level bootstrap sample.

SLP. In principle, it is possible to also estimate the SLP variance using the second bootstrap layer. However, to reduce computational burden, we instead calculate the variance only in the first-level loop, using the LP variance estimates from the second-level bootstrap:

$$\hat{\Sigma}_{b_1}^{SLP} = (\tilde{X}'\tilde{X} + \tilde{\lambda}P)^{-1}(\tilde{X}'\tilde{X})\,\hat{\Omega}_{b_1}^{LP}\,(\tilde{X}'\tilde{X})(\tilde{X}'\tilde{X} + \tilde{\lambda}P)^{-1}, \tag{26}$$

where $\hat{\Omega}_{b_1}^{LP}$ is the LP variance-covariance matrix across all horizons, estimated by stacking second level bootstrap draws of LP and taking covariance of the resulting matrix, and $\tilde{\lambda}$ is fixed at its original sample estimate. This approach avoids the B_2 second-level iterations for SLP while still using the double-bootstrap variance estimates for LP. The result is an $H \times H$ covariance matrix; the diagonal element $[\hat{\Sigma}_{b_1}^{SLP}]_{hh}$ gives the variance at horizon h.

Constructing t-statistics. For all estimators $\mu \in \{LP, VAR, SLP, TLP\}$, we construct t-statistics by centering around the bootstrap mean:

$$t_{h,b_1}^{\mu} = \frac{\hat{\beta}_{h,b_1}^{\mu} - \bar{\beta}_h^{\mu}}{\sqrt{\hat{\Sigma}_{h,b_1}^{\mu}}}, \quad \text{where} \quad \bar{\beta}_h^{\mu} = \frac{1}{B_1}\sum_{b_1=1}^{B_1} \hat{\beta}_{h,b_1}^{\mu}. \tag{27}$$

Subtracting the bootstrap mean rather than the VAR-implied impulse response removes finite sample bias in the numerator of the t-statistics.

Symmetric confidence intervals. The VAR bias under misspecification shifts the location of its confidence interval and leads to asymmetric t-statistic distributions. TLP inherits some of this asymmetry by construction. Additionally, the exponentiation in VAR exacerbates the issues. Symmetric confidence intervals (Hall, 1988) alleviate this problem and, as discussed in Hall (1988), tend to exhibit faster convergence rates than their asymmetric counterparts.

For any estimator $\mu \in \{LP, VAR, SLP, TLP\}$, we rank the bootstrap t-statistics by absolute value and take the $(1-\alpha)$-quantile of $|t_{h,b_1}^{\mu}|$ as the critical value $\hat{t}_{h,1-\alpha}^{\mu}$. The confidence interval is:

$$\left(\hat{\beta}_h^{\mu} - \hat{t}_{h,1-\alpha}^{\mu}\sqrt{\hat{\Sigma}_{h,1}^{\mu}}, \quad \hat{\beta}_h^{\mu} + \hat{t}_{h,1-\alpha}^{\mu}\sqrt{\hat{\Sigma}_{h,1}^{\mu}}\right), \tag{28}$$

where $\hat{\Sigma}^{\mu}_{h,1}$ is the variance estimate from the original sample.

5 Simulation Study

We conduct 1,000 repetitions for sample sizes $T \in \{200, 800\}$, comparing the performance of LP, SLP, TLP, VAR, and BLP estimators. The MSDB uses $B_1 = 200$ first level replications and $B_2 = 100$ second-level replications. The LP, SLP, and TLP estimators are implemented using $p = 10$ lags on the right-hand side, whereas BLP and VAR, as well as the target model for TLP, are specified with either $q \in \{1, 8\}$ lags. The target coverage level is set at 90%. The BLP method utilizes one-sixth of the available data to construct the presample VAR prior, and the coverage is assessed using the posterior distribution.

VARMA(1,∞). The data is generated by the following model

$$Y_t = AY_{t-1} + \Gamma(I + \eta_T \alpha(L))\epsilon_t, \tag{29}$$

where

$$\Gamma^{-1} = \begin{bmatrix} 1 & 0 \\ -1.07 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 0.2 & 0 \\ -0.764 & 0.985 \end{bmatrix}, \tag{30}$$

and $\epsilon_t \sim$ i.i.d. $\mathcal{N}(0, I_2)$. This DGP is taken from Montiel Olea et al. (2026), which simulates the Smets and Wouters (2007) model, a well-known DSGE model that is rich enough to match the second-moment properties of standard macroeconomic time-series data. This model implies a VARMA representation with interpretable structural shocks, implying that any finite order VAR is misspecified. The difference from the DGP in Montiel Olea et al. (2026) is that we use the scaling $\eta_T = \eta \times \sqrt{\frac{200}{T}}$ to check performance of the double bootstrap as T increases. The multiplication by $\sqrt{200}$ is made such that at $T = 200$ we estimate the real DGP when $\eta = 1$. The data generating process used is VARMA$(1, 100)$, with $\eta \in \{1, 2, 4, 8, 16, 32, 64\}$.

5.1 Comparing bootstrap variants

Here, we set $\eta = 1$. Figure 1 compares the MSDB coverage properties against an alternative double bootstrap across four estimators: LP, VAR, SLP, and TLP for $\eta = 1$. The red solid lines represent MSDB, while the purple lines show the alternative which is the same double bootstrap but where the t-statistics used in the studentization are centered around the original VAR estimates, as in Montiel Olea and Plagborg-Møller (2021). LP has 10 lags, and SVAR and the target for TLP have $q \in \{1, 8\}$ lags.

LP and SLP. For LP, both bootstrap methods achieve coverage close to or above the nominal 90% level across all horizons and sample sizes, with no difference in the average confidence interval length. The choice of q changes the coverage and the interval length only marginally. For SLP, there is a drop in coverage at the first few horizons. Nevertheless, at higher horizons the centering around the bootstrap mean yields closer to nominal coverage.

VAR and TLP. For both VAR and TLP, MSDB yields a clear improvement: centering around the VAR estimates leads to substantial undercoverage, consistent with findings in Montiel Olea et al. (2026). MSDB almost restores nominal coverage for both estimators at $T = 200$ and $T = 800$ for $q = 8$. For $q = 1$, there is significant undercoverage for VAR and some undercoverage for TLP. Therefore, it is important to use enough lags in the target VAR.

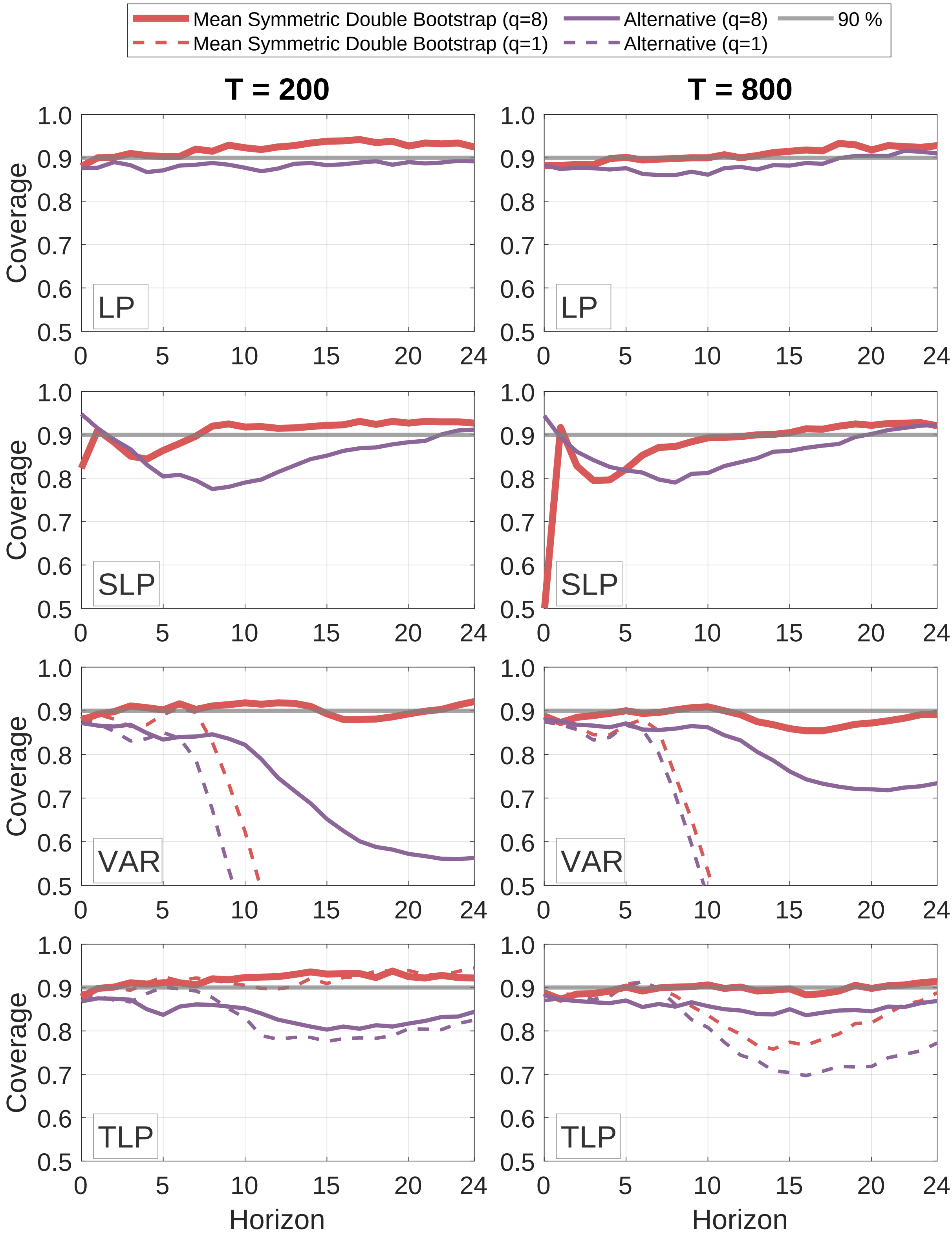

Figure 1: Coverage for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 1$. Red solid line: MSDB with $q = 8$. Purple solid line (alternative method): symmetric double bootstrap centering around VAR estimates. Dashed lines for $q = 1$.

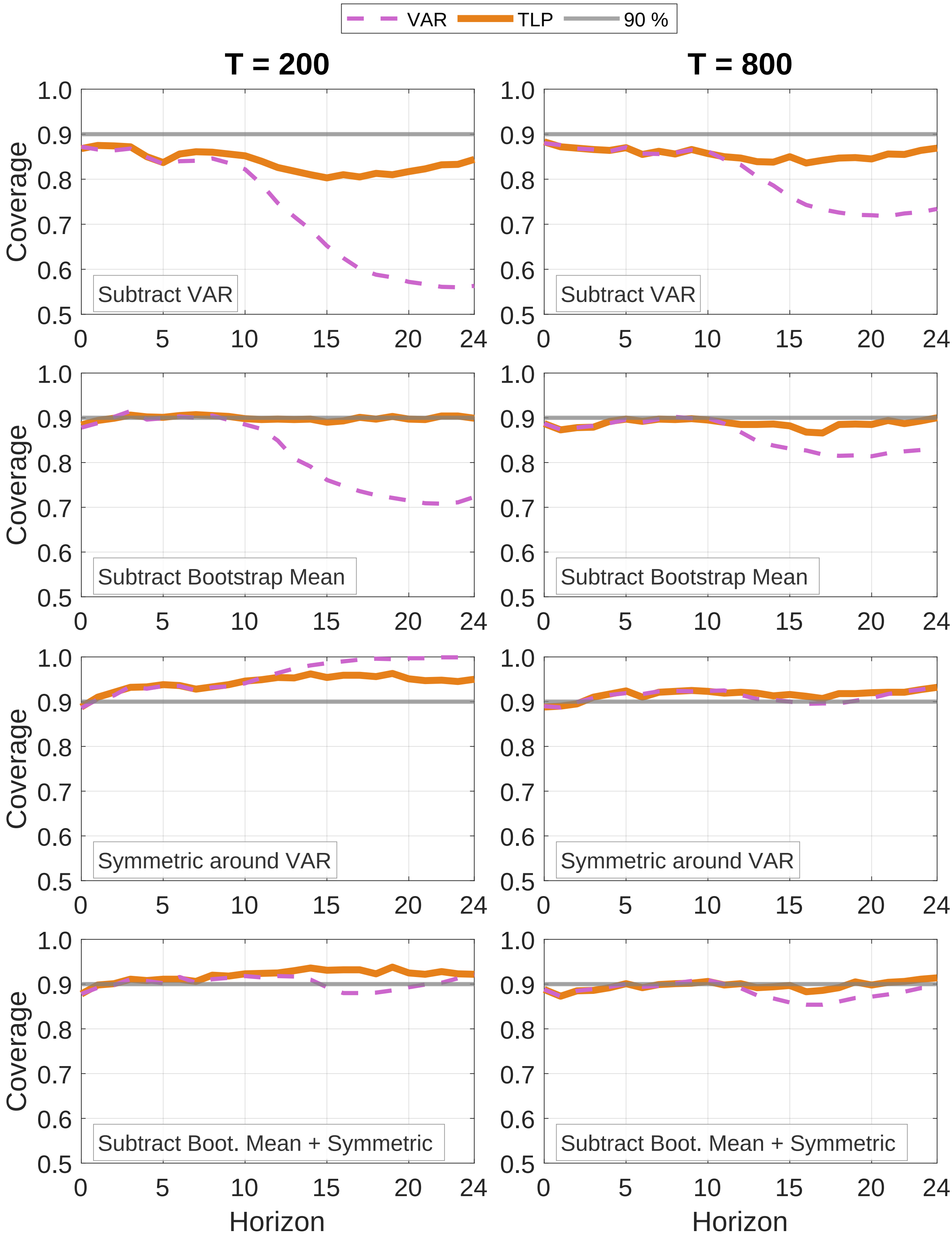

Figure 2: Coverage for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 1$, $q = 8$, Orange solid line: TLP. Pink dashed line: VAR. Rows compare four studentized double bootstraps: Subtract VAR estimate, Subtract Bootstrap Mean, Symmetric around VAR, and Subtract Bootstrap Mean + Symmetric (MSDB).

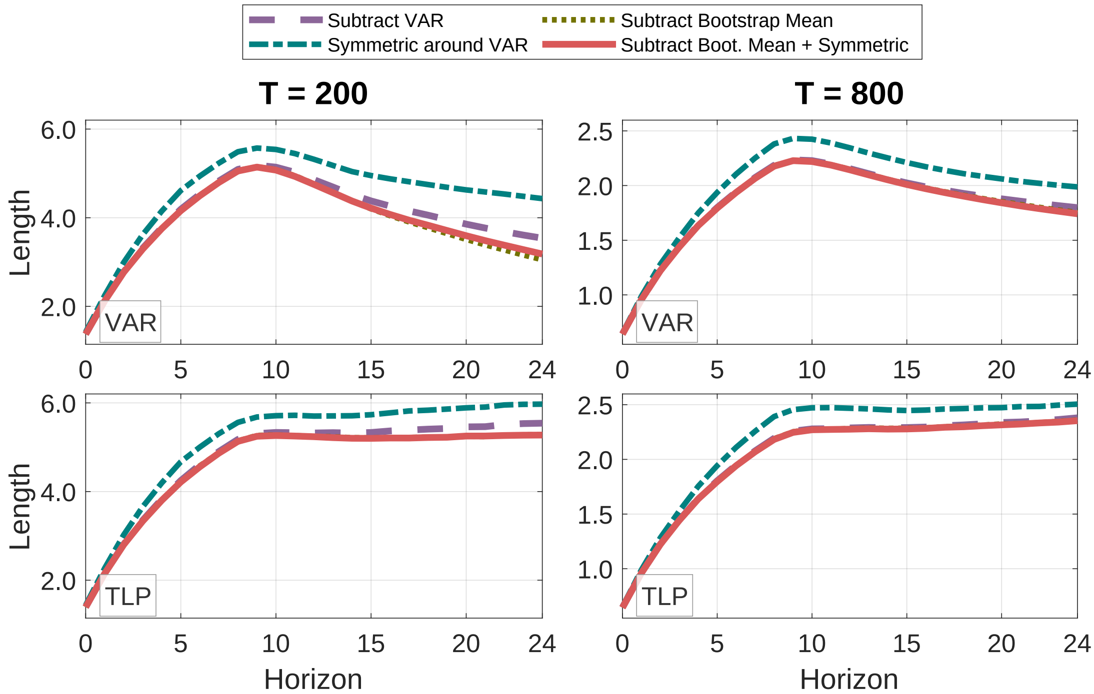

Figure 3: MC Average confidence interval length for VAR and TLP, $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 1$, $q = 8$. Each panel compares four studentized double bootstraps: Subtract VAR estimate, Subtract Bootstrap Mean, Symmetric around VAR, and Subtract Bootstrap Mean + Symmetric (MSDB).

Figure 2 focuses on TLP and VAR with $q = 8$, building up MSDB one change at a time. The first row shows significant undercoverage when using a standard double bootstrap centered at the VAR estimate. The second row centers the t-statistic numerator around the first level bootstrap mean instead, showing substantial improvements in coverage. The third row keeps VAR centering but switches to symmetric intervals, which pushes coverage, especially for VAR, above the nominal level, while increasing the length of the confidence intervals, as seen in Figure 3. The fourth row shows the MSDB, which delivers close to nominal coverage for TLP and less distortion for VAR, with the length of the confidence interval remaining approximately the same as in the asymmetric confidence interval cases.

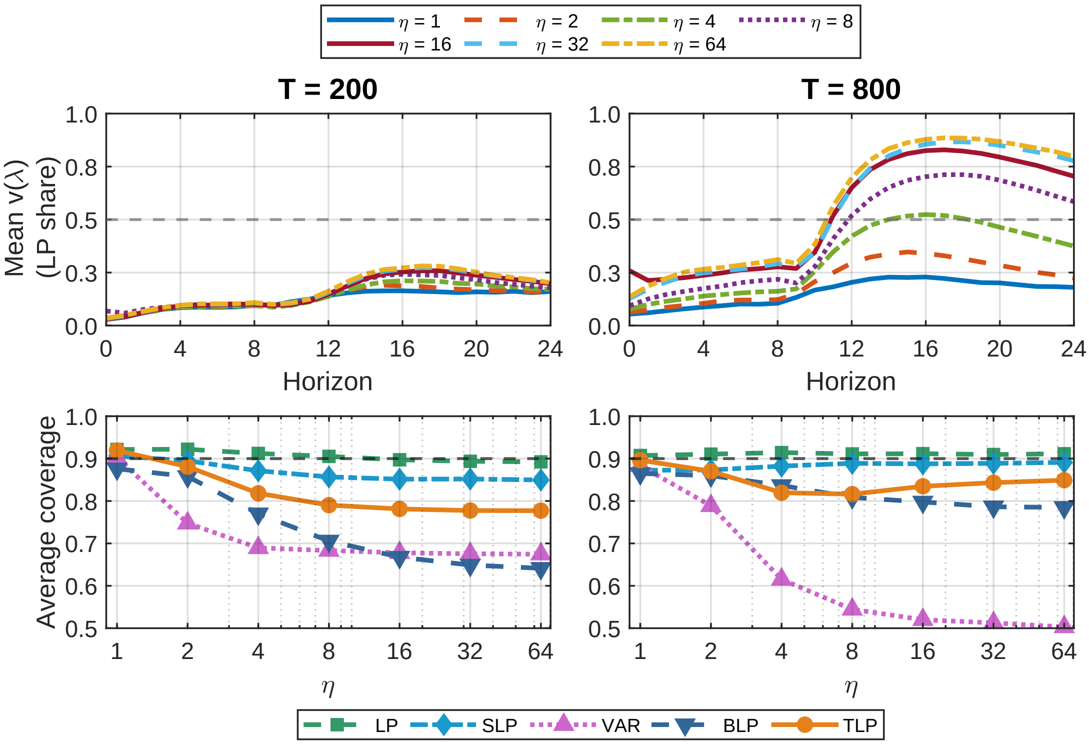

Figure 4: Top row: MC average weight depending on the size of the misspecification η. Bottom row: MC average coverage depending on the size of the misspecification η. $T = 200$ (left) and $T = 800$ (right).

5.2 Comparison Across Estimators

Figure 4 shows the mean - over 24 horizons - TLP weights and mean coverage attributed to LP across different misspecification magnitudes. As expected, the larger the misspecification, the more weight TLP places on LP, which means that first the TLP coverage deteriorates as the misspecification increases, and then it improves again as TLP approaches LP.

Figures 5-7 and Appendix Figures C1-C3 show coverage, length, bias, standard deviation, and RMSE for all five estimators, across different misspecification sizes, horizon by horizon. The coverage and length are for MSDB confidence intervals, while the bias, standard deviation and RMSE are Monte Carlo averages computed with the original sample estimates.

VAR displays significant negative bias at longer horizons while LP remains approximately unbiased for large samples until the misspecification reaches $\eta = 8$. TLP inherits some bias from its VAR target, but this cost is offset by a marked reduction in variance compared to LP. We note that for $\eta = 1$, the misspecification is small enough so that the bias - variance trade-off is not present, and VAR has the smallest RMSE. However, even in this case, for most horizons, the RMSE of TLP is similar, and the VAR exhibits slight coverage distortions while the coverage of TLP is close to nominal. As η increases, the trade-off becomes visible, and TLP exhibits smaller RMSE than VAR, with marked RMSE improvements over VAR at $\eta = \{2, 4\}$. For the latter misspecifications, the TLP bias is markedly smaller than that of the VAR, and the variance is smaller than that of the LP, no longer increasing with the horizon. As the misspecification increases, some coverage distortions of TLP are to be expected, as discussed in the introduction; however, they are markedly smaller than those of the VAR. Interestingly, as shown in Figure 4, these coverage distortions become smaller again as the misspecification further increases and more weight is placed on LP.

Since we allow for conditional heteroskedasticity, we also simulate the VARMA(1,100) with $\eta = 200^{1/2} \times T^{-1/2}$ and the errors following a GARCH(1,1) process: $\epsilon_t = \sigma_t z_t$ with $z_t \sim$ i.i.d.$N(0, 1)$ and $\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, where $\omega = 0.05$, $\alpha = 0.10$, and $\beta = 0.85$. The results are similar to Figures 5-7 and are shown in Figures C4 - C6 in Appendix C.2.

SLP and BLP. SLP achieves nominal coverage at longer horizons but suffers from severe undercoverage at short horizons. This pattern is expected, as smoothing requires neighboring estimates, which are unavailable at the initial horizon. Even when coverage is near nominal, the average length of the CI does not decrease meaningfully compared to standard LP. It also exhibits substantial bias at smaller horizons without meaningful reductions in variance relative to LP. BLP also suffers from substantial "undercoverage" at short horizons, with wide credible intervals, likely because it uses a sixth of the sample as a presample to initialize the prior distribution, reducing effective sample size.

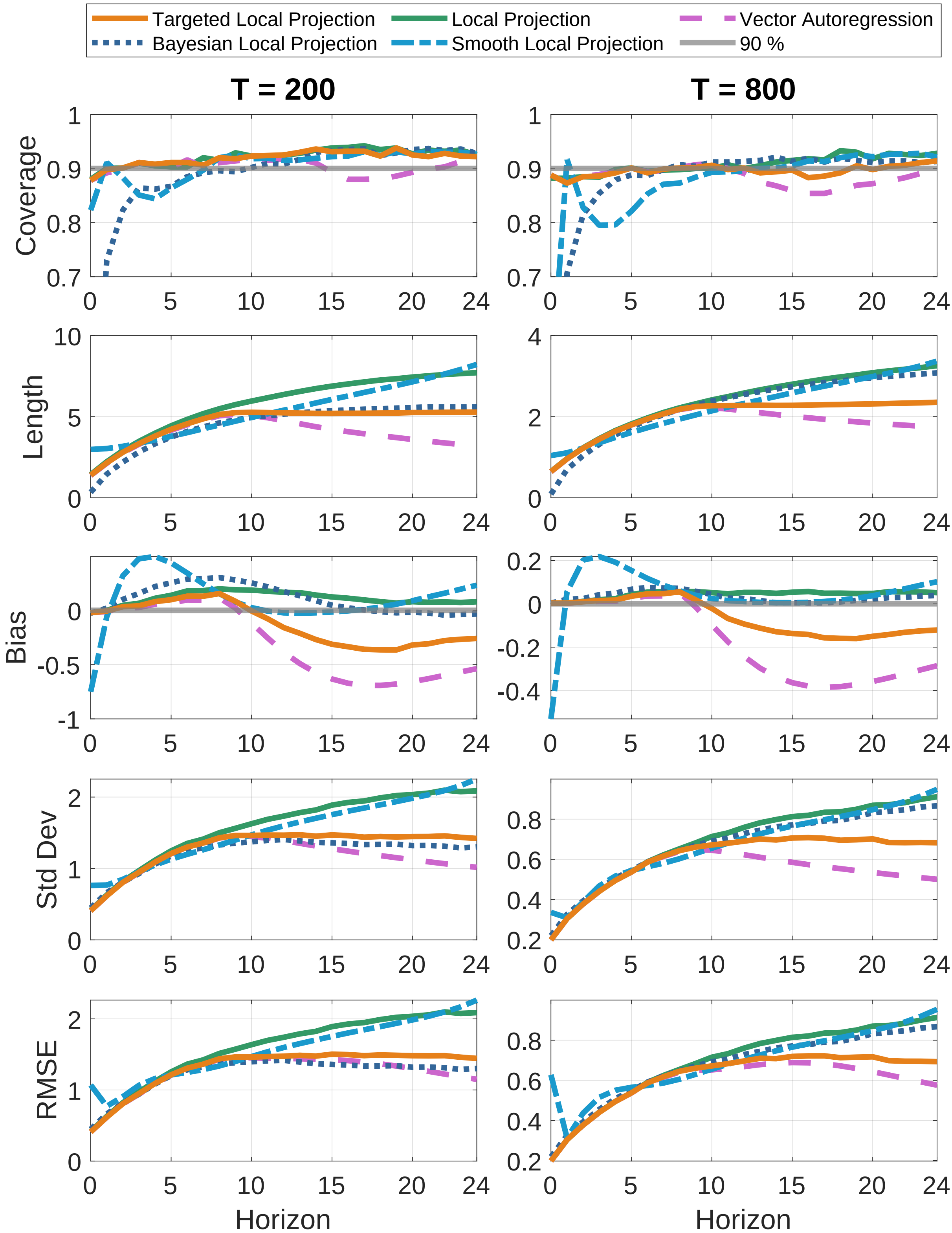

Figure 5: Coverage, length, bias, standard deviation, and RMSE for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 1$, conditionally homoskedastic errors. Methods: Targeted Local Projections (10,8), Local Projections (10), Vector Autoregression (8), Smooth Local Projections (10) and Bayesian Local Projections (8,8). Inference by MSDB for TLP, LP, VAR and SLP.

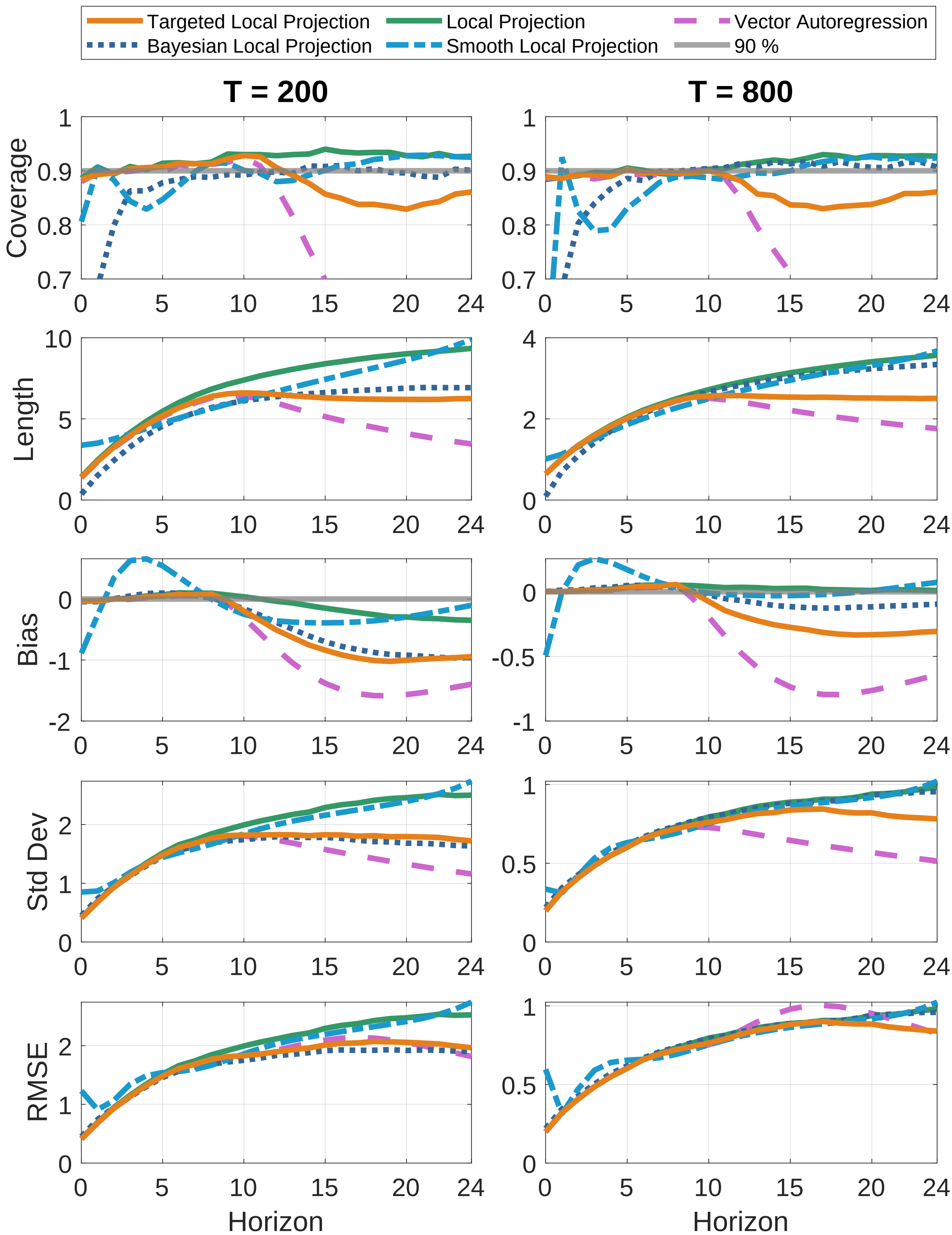

Figure 6: Coverage, length, bias, standard deviation, and RMSE for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 2$, conditionally homoskedastic errors. Methods: Targeted Local Projections (10,8), Local Projections (10), Vector Autoregression (8), Smooth Local Projections (10) and Bayesian Local Projections (8,8). Inference by MSDB for TLP, LP, VAR and SLP.

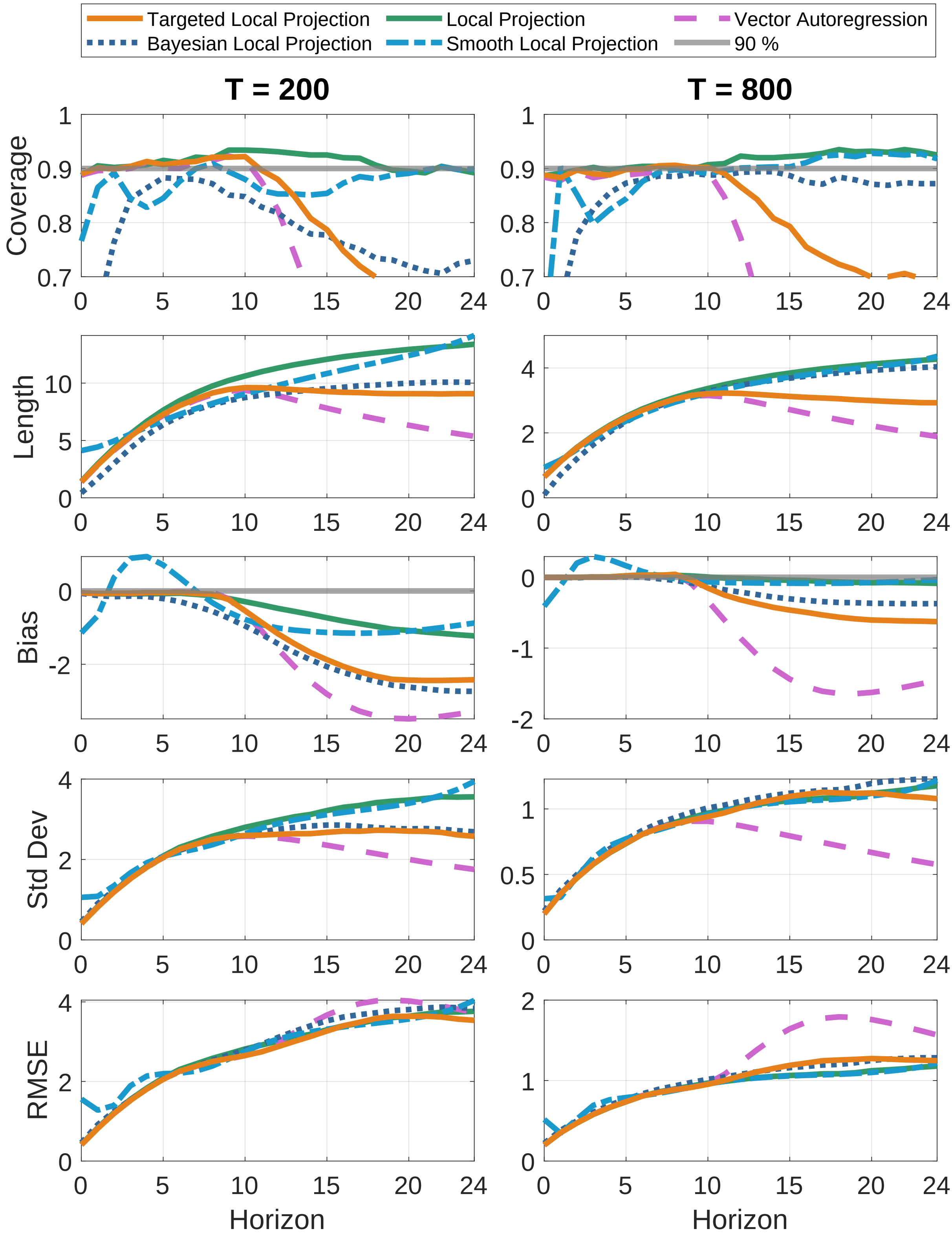

Figure 7: Coverage, length, bias, standard deviation, and RMSE for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 4$, conditionally homoskedastic errors. Methods: Targeted Local Projections (10,8), Local Projections (10), Vector Autoregression (8), Smooth Local Projections (10) and Bayesian Local Projections (8,8). Inference by MSDB for TLP, LP, VAR and SLP.

6 Empirical Application

Following Ferreira et al. (2023), we estimate impulse response functions of key macroeconomic variables to a federal funds rate shock using a seven-variable system that includes real GDP, real consumption, real investment, total hours worked, real wages, the GDP deflator, and the federal funds rate. The data are from the US and span 1954Q3 to 2019Q4, with all variables expressed in logarithms and seasonally adjusted at the annual rate, except for the policy rate. More details on the data construction can be found in Ferreira et al. (2023), Online Appendix A.

As in Ferreira et al. (2023), the monetary policy shock is identified via a recursive ordering, with the federal funds rate placed last. We estimate TLP, LP, and VAR impulse responses with $(p, q) = (8, 4)$ and $(p, q) = (10, 4)$ using the MSDB for inference. The estimated impulse responses are shown in Figure 8 for $p = 10$, and in Appendix Figure D1 for $p = 8$. As expected, following a contractionary monetary policy shock, real GDP, consumption, investment, and hours worked decline. From the two figures, it is visible that for most variables except hours worked, the GDP deflator and the federal funds rate, the LP and the VAR estimates are quite different at higher horizons, and a possible reason for these differences could be that the VAR is misspecified at higher horizons, as also suggested in Ferreira et al. (2023). When the two estimates are very different, the TLP estimates place more weight on LP, and when they are similar, more weight is placed on the VAR.

The TLP confidence intervals are tighter than their LP counterparts. Tables 1-2 show the horizons at which the impulse responses are statistically significant at the 90% level for these first four variables, with blue highlighting the horizons at which TLP estimates are significant while LP estimates are not. We see that particularly for real investment, the TLP estimates are larger and remain significant for most horizons, demonstrating the usefulness of our estimator for empirical applications in reducing variance.

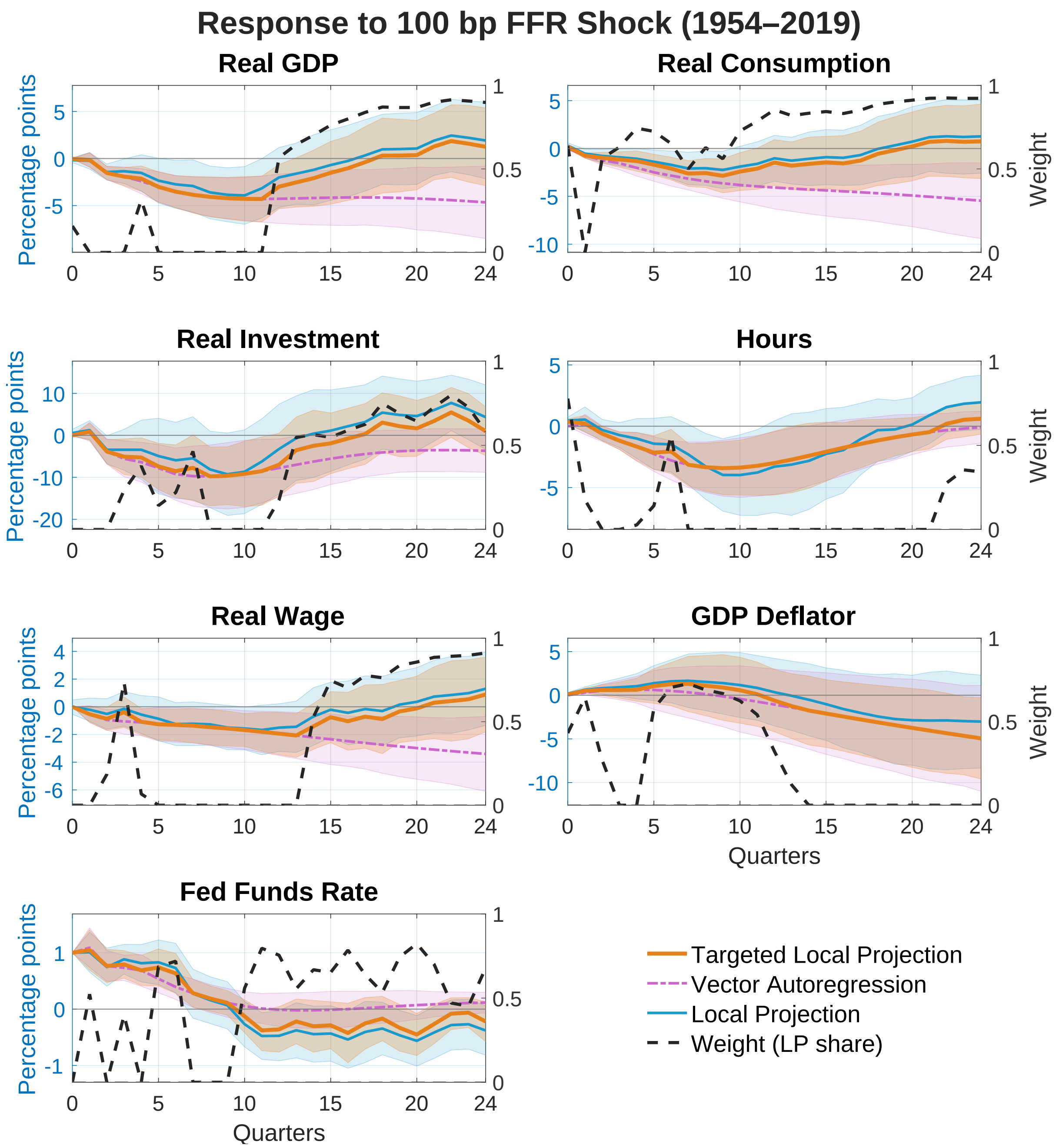

Figure 8: Impulse responses to a 100 basis point shock to the federal funds rate (1954–2019). The panels display the responses of Real GDP, Real Consumption, Real Investment, Hours, Real Wage, GDP Deflator, and the federal funds rate. Shaded regions correspond to 90% confidence intervals. Estimation uses $p = 10$ and $q = 4$ lags.

Horizon	Real GDP			Real Consumption			Real Investment			Hours Worked		
h	LP	TLP	VAR	LP	TLP	VAR	LP	TLP	VAR	LP	TLP	VAR
0	−0.25*	−0.16	0.00	0.31*	0.22	0.00	0.63	0.12	0.00	0.61*	0.54*	0.00
1	−0.46	−0.27	−0.15	−0.47*	−0.65*	−0.74*	1.21	0.88	0.88	0.72	0.51	0.14
2	−1.77*	−1.54*	−1.54*	−0.78*	−0.93*	−1.23*	−3.48	**−3.85***	−3.85*	−0.03	−0.32	−0.61
3	−1.67*	−1.92*	−1.92*	−0.89*	−1.06*	−1.54*	−3.27	−4.47	−5.58*	−0.34	−0.61	−1.18*
4	−1.79	**−2.41***	−2.41*	−1.10*	−1.30*	−2.02*	−2.88	−4.13	−6.40*	−0.56	−0.88	−1.68*
5	−2.47*	−3.02*	−3.02*	−1.31*	−1.54*	−2.49*	−4.08	−5.65	−7.82*	−0.99	−1.38	−2.27*
6	−2.63*	−3.48*	−3.52*	−1.56*	−1.84*	−2.83*	−4.98	−6.62	−9.26*	−1.26	−1.67	−2.79*
7	−2.69	**−3.47***	−3.85*	−1.82*	−2.16*	−3.15*	−4.93	−6.61	−9.73*	−2.08	**−2.76***	−3.14*
8	−3.11*	−4.08*	−4.08*	−1.73*	−2.07*	−3.43*	−6.81	**−9.79***	−9.79*	−3.01*	−3.35*	−3.35*
9	−3.14*	−4.22*	−4.22*	−1.84*	−2.23*	−3.64*	−6.84	**−9.63***	−9.63*	−3.42*	−3.42*	−3.42*
10	−3.15*	−4.29*	−4.29*	−1.64	**−2.02***	−3.82*	−5.72	**−9.10***	−9.18*	−3.27*	−3.38*	−3.38*
11	−2.65	**−3.65***	−4.30*	−1.38	−1.76	−3.96*	−4.06	−6.95	−8.53*	−3.25	**−3.23***	−3.23*
12	−1.94	**−2.74***	−4.28*	−0.97	−1.33	−4.08*	−2.27	−4.85	−7.79*	−3.03	**−3.00***	−3.00*
13	−1.71	**−2.53***	−4.24*	−1.31	−1.75	−4.19*	−0.01	−2.23	−7.01*	−3.07	**−2.72***	−2.72*
14	−1.41	−2.22	−4.20*	−1.08	−1.52	−4.28*	0.56	−1.86	−6.26	−3.06	−2.41	−2.41
15	−0.98	−1.74	−4.16*	−0.93	−1.39	−4.38*	1.13	−1.46	−5.56	−2.63	−2.08	−2.08
16	−0.42	−1.11	−4.14*	−0.77	−1.23	−4.47*	2.42	−0.00	−4.96	−2.31	−1.76	−1.76

Table 1: Impulse Responses to 100 bp Federal Funds Rate Shock (bold blue indicates TLP(8,4) significance where LP(8) is insignificant; a star indicates 90% significance). The table is truncated to include the first 16 quarters.

Horizon	Real GDP			Real Consumption			Real Investment			Hours Worked		
h	LP	TLP	VAR	LP	TLP	VAR	LP	TLP	VAR	LP	TLP	VAR
0	−0.17	−0.03	0.00	0.29	0.19	0.00	0.55	0.00	0.00	0.47*	0.36*	0.00
1	−0.24	−0.15	−0.15	−0.53*	−0.74*	−0.74*	1.24	0.88	0.88	0.55	0.21	0.14
2	−1.44*	−1.54*	−1.54*	−0.78*	−0.97*	−1.23*	−3.44*	−3.85*	−3.85*	−0.31	**−0.61***	−0.61
3	−1.35*	−1.92*	−1.92*	−0.92*	−1.15*	−1.54*	−3.43	**−5.08***	−5.58*	−0.72	**−1.18***	−1.18*
4	−1.53	**−2.14***	−2.41*	−1.07*	−1.31*	−2.02*	−3.44	**−5.28***	−6.40*	−1.01	**−1.66***	−1.68*
5	−2.35	**−3.02***	−3.02*	−1.38*	−1.69*	−2.49*	−4.93	**−7.41***	−7.82*	−1.43	**−2.15***	−2.27*
6	−2.75*	−3.52*	−3.52*	−1.70*	−2.09*	−2.83*	−5.95	**−8.54***	−9.26*	−1.48	**−2.06***	−2.79*
7	−2.92*	−3.85*	−3.85*	−2.09*	−2.62*	−3.15*	−5.46	−7.75	−9.73*	−2.30	**−3.14***	−3.14*
8	−3.61*	−4.08*	−4.08*	−2.06*	−2.57*	−3.43*	−8.14	**−9.79***	−9.79*	−3.27*	−3.35*	−3.35*
9	−3.85*	−4.22*	−4.22*	−2.24*	−2.85*	−3.64*	−9.28	**−9.63***	−9.63*	−3.97*	−3.42*	−3.42*
10	−3.92*	−4.29*	−4.29*	−1.89	**−2.41***	−3.82*	−8.67	**−9.18***	−9.18*	−3.98*	−3.38*	−3.38*
11	−3.17*	−4.30*	−4.30*	−1.60	−2.11	−3.96*	−6.25	**−8.53***	−8.53*	−3.78*	−3.23*	−3.23*
12	−2.01	**−2.99***	−4.28*	−1.02	−1.47	−4.08*	−3.26	−6.99	−7.79*	−3.30	**−3.00***	−3.00*
13	−1.61	−2.54	−4.24*	−1.27	−1.79	−4.19*	−0.67	−3.54	−7.01*	−3.13	**−2.72***	−2.72*
14	−1.20	−2.10	−4.20*	−1.08	−1.61	−4.28*	0.41	−2.50	−6.26	−2.83	−2.41	−2.41
15	−0.69	−1.52	−4.16*	−0.91	−1.45	−4.38*	1.12	−1.91	−5.56	−2.26	−2.08	−2.08
16	−0.26	−1.04	−4.14*	−0.97	−1.56	−4.47*	2.17	−0.76	−4.96	−1.95	−1.76	−1.76

Table 2: Impulse Responses to 100 bp Federal Funds Rate Shock (bold blue indicates TLP(10,4) significance where LP(10) is insignificant; a star indicates 90% significance). The table is truncated to include the first 16 quarters.

7 Conclusion

This paper introduces targeted local projections, a frequentist shrinkage estimator that averages across LP and VAR impulse responses with data-driven, horizon-specific weights. By shrinking LP toward VAR at each horizon, TLP reduces variance at longer horizons where LP estimates become noisy, while the optimal weights ensure that the estimator places more weight on the asymptotically unbiased LP when the discrepancy between estimators is large. The shrinkage parameter has a closed-form solution, making implementation straightforward. For inference, we propose the Mean Symmetric Double Bootstrap, which improves coverage relative to the standard single bootstrap procedures.

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Data availability and AI usage. Code and data are available at https://github.com/anemtyrev/Targeted_Local_Projections. AI tools used: Claude, Claude Code, ChatGPT, Codex. AI usage: coding, proofreading.

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A The Importance of Projecting Out Controls in Regularized Local Projections

All estimators discussed in this paper—$\hat{\beta}_h^{LP}$, $\hat{\beta}_h^{TLP}$, $\hat{\beta}_h^{SLP}$, and $\hat{\beta}_h^{VAR}$—are scalars. This is achieved by projecting out control variables W prior to estimation. Without this step, regularization becomes problematic for two reasons.

First, if we apply uniform regularization to all parameters including controls, substantial further derivations are required to identify the target for the control coefficients.

Second, if we regularize only the parameter of interest while setting zero targets for controls, the objective function for TLP becomes

$$Q_n(\theta) = \|\check{Y}_h - \theta_h \check{X}\|^2 + \tilde{\lambda}\|\theta_h - \hat{\theta}_h^{VAR}\|_\Lambda^2 \tag{31}$$

where $\check{X} = [Y_2\ W]$, $\check{Y}_h = Y_{1,h}$, and

$$\hat{\theta}_h^{VAR} = \begin{bmatrix} \hat{\beta}_h^{VAR} \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(pk+1)\times 1}, \quad \Lambda = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}_{(pk+1)\times(pk+1)}. \tag{32}$$

The solution is

$$\hat{\theta}_h^{\text{TLP}} = \phi(\tilde{\lambda})\hat{\theta}_h^{LP} + (I_{pk+1} - \phi(\tilde{\lambda}))\hat{\theta}_h^{VAR}, \tag{33}$$

where $\phi(\tilde{\lambda}) = (\check{X}'\check{X} + \tilde{\lambda}\Lambda)^{-1}(\check{X}'\check{X})$. The matrix $\phi(\tilde{\lambda})$ is not diagonal, so any $\tilde{\lambda} \neq 0$ introduces bias into the control coefficients.

Yamada (2017) shows that projecting out controls preserves both the parameter of interest and its variance, up to a scaling factor, in ridge regression (but not in generalized ridge regression). This approach eliminates the targeting ambiguity and prevents cross-contamination between the parameter of interest and controls. For this reason, projecting out controls is essential for both TLP and SLP estimators.

B Proofs

Lemma 1. Under Assumptions 1 and 2, y_t, $u_t = \Gamma(\epsilon_t + T^{-\zeta}\alpha(L)\epsilon_t)$, and $u_t - \Gamma\epsilon_t$ are $\mathcal{L}_d$ near-epoch dependent (NED) of size $m^{-1/2}$ on the α-mixing process ϵ_t of size $-r/(r-2)$, where recall $d = 4 + 2\delta$, and $r = 2 + \delta = d/2$.

Proof of Lemma 1. Under Assumption 1, y_t can be inverted into a VMA(∞). Letting $\alpha(L) = \sum_{\ell=1}^{\infty} \alpha_\ell L^\ell$, with the convention $\alpha_0 = 0$,

$$y_t = \sum_{j=0}^{\infty}(A^j\Gamma + T^{-\zeta}A^j\Gamma\alpha_j)\epsilon_{t-j} \equiv \sum_{j=0}^{\infty} B_{T,j}\epsilon_{t-j}.$$

By Assumption 1, $\sup_t E\|\epsilon_t\|_d < M$, for a generic constant M - which we use throughout to denote various uniform bounds - and the VMA coefficients are absolutely summable $\sup_T \sum_{j=0}^{\infty} \|B_{T,j}\| < \infty$, where $\|\cdot\|$ denotes the Euclidean norm for vectors, and the sup-norm for matrices. Since $E(y_t|\mathcal{F}_{t-m}^{t+m}) = E(y_t|\mathcal{F}_{t-m}) = \sum_{j=0}^{m} B_{T,j}\epsilon_{t-j}$,

$$\begin{aligned}\|y_t - E(y_t|\mathcal{F}_{t-m}^{t+m})\|_d &= \left(\mathbb{E}\left\|\sum_{j=m+1}^{\infty} B_{T,j}\epsilon_{t-j}\right\|^d\right)^{1/d} \leq \left(\sum_{j=m+1}^{\infty} \|B_{T,j}\|^d \mathbb{E}\|\epsilon_{t-j}\|^d\right)^{1/d} \\ &\leq \left(M\sum_{j=m+1}^{\infty} \|B_{T,j}\|^d\right)^{1/d} = M^{1/d}\left(\sum_{j=m+1}^{\infty} \|B_{T,j}\|^d\right)^{1/d}.\end{aligned}$$

Since $||B_{T,j}||$ are absolutely summable, they are also $d > 4$ summable. Moreover, $\sum_{j=m+1}^{\infty} \|B_{T,j}\|^d \leq \|\Gamma\|^d \left(\sum_{j=m+1}^{\infty} \|A^{jd}\| \times \|1 + \alpha_j T^{-\zeta}\|^d\right) \leq M\left(\sum_{j=m+1}^{\infty} \|A^j\|^d\right)$. Since $\sum_{j=m+1}^{\infty} \|A^j\|^d \to 0$ as $m \to \infty$, y_t is $\mathcal{L}_d$ NED on ϵ_t, with approximation constants $c_t = M$ and $\nu_{m,1} = \left(\sum_{j=m+1}^{\infty} \|B_{T,j}\|^d\right)^{1/d}$. Since A^m decays faster than any polynomial rate, y_t is $\mathcal{L}_d$ NED of every finite polynomial size, therefore also of size m^{-1} and $m^{-1/2}$. The result for u_t can be obtained following the same steps as for y_t but without needing the VMA(∞) inversion. Noting that $E(u_t|\mathcal{F}_{t-m}^{t+m}) = E(u_t|\mathcal{F}_{t-m}) = \sum_{j=0}^{m} \Gamma(1 + T^{-\zeta})\alpha_j\epsilon_{t-j}$, we have:

$$\|u_t - E(u_t|\mathcal{F}_{t-m}^{t+m})\|_d = \left(\mathbb{E}\left\|\sum_{j=m+1}^{\infty}\Gamma(1+T^{-\zeta})\alpha_j\epsilon_{t-j}\right\|^d\right)^{1/d}$$
$$\leq \left(\sum_{j=m+1}^{\infty}\|\Gamma(1+T^{-\zeta})\alpha_j\|^d\mathbb{E}\|\epsilon_{t-j}\|^d\right)^{1/d} \leq M\left(\sum_{j=m+1}^{\infty}\|\alpha_j\|^d\right)^{1/d}$$
$$\equiv M\nu_{m,2}.$$

Therefore, u_t is $\mathcal{L}_d$ NED on the α-mixing process ϵ_t, with approximation constants $c_t = M$ and $\nu_{m,2}$ which also decay exponentially by Assumption 1; hence, the NED order of u_t is also m^{-1} or just $m^{-1/2}$. Moreover, $u_t - \Gamma\epsilon_t$ inherits the dependence properties of u_t by similar derivations as above. □

Lemma 2 (WLLN). If $\{h_t\}$ is a mean zero random vector and $\mathcal{L}_2$ NED on an α-mixing process v_t of size $\nu_m = m^{-1/2}$ with constants $c_t = M$ fixed for a generic M, with $\sup_t \mathbb{E}|v_{t,i}|^{2+\delta} < M < \infty$, where $v_{t,i}$ is the i^{th} element of v_t, then $T^{-1}\sum_{t=1}^{T} h_t \xrightarrow{p} 0$.

Proof of Lemma 2. From Proposition 2.9 in Wooldridge and White (1988), h_t is an $\mathcal{L}_2$ mixingale of size $-1/2$ with constants $c_t = M$ and $\psi_m = 5\alpha_{[m/2]}^{1/2-1/d} + \nu[m/2]$. Hence, it is also an $\mathcal{L}_1$ mixingale of the same size with the same constants, and $\lim\sum_{T=1}^{T} c_t = M < \infty$. Hence, by Andrews (1988), Theorem 1, the desired result follows. □

Lemma 3 (CLT). If $\{h_t\}$ is a mean zero random vector and $\mathcal{L}_2$ NED on an α-mixing process v_t of size $-r^*/(r^*+2)$, with $\sup_t \mathbb{E}|v_{t,i}|^{r^*} < M < \infty$ for some $r^* > 2$, where $v_{t,i}$ is the i^{th} element of v_t, and the long-run variance exists and it is positive definite: $\lim_{T\to\infty} \mathrm{Var}\left(T^{-1/2}\sum_{t=1}^{T} h_t\right) = V$, then $T^{-1/2}\sum_{t=1}^{T} h_t \xrightarrow{d} \mathcal{N}(0, V)$.

Proof of Lemma 3. Since h_t is an $\mathcal{L}_1$ mixingale of size $-1/2$ with constants defined as in the proof of Lemma 2, the result follows by applying Theorem 2.11 in Wooldridge and White (1988), as already verified in Hall et al. (2012).

Proof of Theorem 1. The asymptotic bias for the VAR estimator was derived in Montiel Olea et al. (2026) under Assumption 1. For the LP estimator, consider the regression coefficient $\hat{\theta}_h$ on all coefficients, obtained with regression model:

$$y_{i,t+h} = y_{j,t}\beta_h + w_t'\gamma_h + u_{t+h}^h \equiv z_t'\theta_h + u_{t+h}^h,$$

where w_t contains all the p lags of y_t included in LP, $z_t = (y_{j,t}, w_t')'$, $\theta_h = (\beta_h, \gamma_h')'$, γ_h are functions of the original VAR coefficients obtained by backward substituting the model in (1), and u_{t+h}^h are the errors which can be shown, by backward substitution, to be linear combinations of $u_{t+h}, \ldots, u_t$, the reduced-form errors of the VAR. Denote this linear combination by

$$u_{t+h}^h = \sum_{n=0}^{h} \Phi_n' u_{t+h-n} = \sum_{n=0}^{h} \Phi_n' \Gamma \epsilon_{t+h-n} + T^{-\zeta} \sum_{n=0}^{h} \Phi_n' \Gamma \alpha(L) \epsilon_{t+h-n}. \tag{34}$$

Note that

$$\begin{aligned} \hat{\theta}_h^{LP} &= \theta_h + \left(T^{-1} \sum_{t=1}^{T} z_t z_t' \right)^{-1} \left(T^{-1} \sum_{t=1}^{T} z_t u_{t+h}^h \right) \qquad (35) \\ &= \theta_h + \left(T^{-1} \sum_{t=1}^{T} z_t z_t' \right)^{-1} \left(T^{-1/2} \sum_{n=0}^{h} z_t \Phi_n' \Gamma \epsilon_{t+h-n} + T^{-1/2-\zeta} \sum_{n=0}^{h} z_t \Phi_n' \Gamma \alpha(L) \epsilon_{t+h-n}. \right) \end{aligned}$$

Since ϵ_t is MDS by Assumption 1, and z_t either contains lags of y_t or $y_{j,t}$, while Γ is a lower-triangular matrix, $\mathbb{E}(z_t \Phi_n' \Gamma \epsilon_{t+h}) = 0$ for all $h \geq 0$, hence

$$\mathbb{E}\left(T^{-1/2} \sum_{n=0}^{h} z_t \Phi_n' \Gamma \epsilon_{t+h-n} \right) = 0.$$

Let $z_{i,t}$ be the i^{th} element of z_t, and denote by Ψ_{n,ℓ,k^*} the $(k^*)^{th}$ element of $\Psi_{n,\ell} \equiv \Phi_n' \Gamma \alpha_\ell$, where $\alpha(L)=\sum_{\ell=1}^{\infty}\alpha_\ell L^\ell$, and L is the lag operator. Then:

$$\begin{aligned}
T^{-1/2-\zeta}\sum_{n=0}^{h}\mathbb{E}(z_t\Phi_n'\Gamma\alpha(L)\epsilon_{t+h-n}) &\leq \mathbb{E}\left|T^{-1/2-\zeta}\sum_{n=0}^{h}z_{i,t}\sum_{\ell=1}^{\infty}\Psi_{n,\ell}\epsilon_{t+h-n-\ell}\right| \\
&\leq T^{-1/2-\zeta}\sum_{n=0}^{h}\sum_{\ell=1}^{\infty}\mathbb{E}|z_{i,t}\Psi_{n,\ell}\epsilon_{t+h-n-\ell}| \\
&\leq T^{-1/2-\zeta}\sum_{n=0}^{h}\sum_{\ell=1}^{\infty}\mathbb{E}\left|\sum_{k^*=1}^{k}z_{i,t}\Psi_{n,\ell,k^*}\epsilon_{k^*,t+h-n-\ell}\right| \\
&\leq T^{-1/2-\zeta}\sum_{n=0}^{h}\sum_{\ell=1}^{\infty}\sum_{k^*=1}^{k}\times|\Psi_{n,\ell,k^*}|\sup_{t,i}\|z_{i,t}\|_2\sup_{t,k^*}\|\epsilon_{k^*,t}\|_2.
\end{aligned}$$

By Assumption 1 (i), $\sup_{t,k^*}|\epsilon_{k^*,t}|_2<M$. Also,

$$\sup_t\|y_t\|_2\leq\sum_{j=0}^{\infty}\|B_{T,j}\|\times\sup_t\|\epsilon_t\|_2=\sum_{j=0}^{\infty}\|B_{T,j}\|\times\left(\mathbb{E}\left(\sum_{k',k^*=1}^{k}\epsilon_{k',t}\epsilon_{k^*,t}\right)^2\right)^{1/2} \tag{36}$$

$$=\sum_{j=0}^{\infty}\|B_{T,j}\|\times\left(\sum_{k^*=1}^{k}\mathbb{E}(\epsilon_{k^*,t}^2)\right)^{1/2}=\sum_{j=0}^{\infty}\|B_{T,j}\|\times\left(\sum_{k^*=1}^{k}\sigma_{k^*}^2\right)^{1/2}<M, \tag{37}$$

where the first inequality is the triangle inequality, the second and third equalities follow from Assumption 1(i), and the last inequality from Assumption 1 (i), (ii) and (v). Hence, $\sup_t\|y_t\|_2<M$, and

$$\mathbb{E}\left|T^{-1/2-\zeta}\sum_{n=0}^{h}z_{i,t}\sum_{\ell=1}^{\infty}\Psi_{n,\ell}\epsilon_{t+h-n-\ell}\right|\leq T^{-1/2-\zeta}kM\sum_{\ell=1}^{\infty}\max_{k^*}\|\Psi_{n,\ell,k^*}|=o(T^{-1/2}),$$

where the equality follows from $|\Psi_{n,\ell}|$ being absolutely summable. Hence,

$$\mathbb{E}\left(T^{-1}\sum_{t=1}^{T}z_tu_{t+h}^h\right)=o(T^{-1/2}). \tag{38}$$

It remains to show that $T^{-1}\sum_{t=1}^{T}z_tz_t'$ is uniformly bounded. However, for simplicity, we now also verify Lemma 2, WLLN, for $T^{-1}\sum_{t=1}^{T}z_tz_t'$. Because y_t is $\mathcal{L}_d$ NED of size $-1/2$ on ϵ_t by Lemma 1, so is z_t, and so z_tz_t' but with lower norm $\mathcal{L}_r$ NED, by Theorems 17.8-17.10 in Davidson (1994), $r=2+\delta$, hence also with $\mathcal{L}_2$ norm. By Hölder's inequality, $\sup_t E|z_{i,t}z_{i',t}|^{2+\delta}<M$, with $z_{i,t}$ the i^{th} element of z_t. We now show that $\mathbb{E}(y_ty_{t-\ell}')=$

$E(\tilde{y}_t\tilde{y}_{t-\ell}) + o(1)$. Since $y_t = \tilde{y}_t + T^{-\zeta}\bar{u}_t$, where $\bar{u}_t = \Gamma\alpha(L)\epsilon_t$, and recall that $\tilde{y}_t$ is the process in (1) purged of misspecification, hence setting $\zeta = 0$. Then we have:

$$\begin{aligned}\mathbb{E}(y_t y_{t-\ell}) &= \mathbb{E}[(\tilde{y}_t + T^{-\zeta}\bar{u}_t)(\tilde{y}_{t-\ell} + T^{-\zeta}\bar{u}_{t-\ell})] \\ &= \mathbb{E}(\tilde{y}_t\tilde{y}_{t-\ell}) + T^{-\zeta}\mathbb{E}(y_t\bar{u}_{t-\ell}) + T^{-\zeta}\mathbb{E}(y_{t-\ell}\bar{u}_t) + T^{-2\zeta}\mathbb{E}(y_{t-\ell}\bar{u}_{t-\ell}).\end{aligned}$$

By similar arguments as for $z_t\Phi_n'\Gamma\epsilon_{t+h-n}$, $\sup_{t,\ell}\mathbb{E}\|y_{t-\ell}\bar{u}_t\| < M$, $\sup_{t,\ell}\mathbb{E}\|y_t\bar{u}_{t-\ell}\| < M$, and $\sup_{t,\ell}\mathbb{E}\|y_{t-\ell}\bar{u}_{t-\ell}\| < M$. Hence, $\mathbb{E}(y_t y_{t-\ell}) = \mathbb{E}(\tilde{y}_t\tilde{y}_{t-\ell}) + o(1)$, and because z_t contains $y_{j,t}$ and lags of y_t,

$$\mathbb{E}(z_t z_t') = \mathbb{E}(\tilde{z}_t\tilde{z}_t') + o(1) \equiv \tilde{S}_q + o(1),$$

where $\tilde{z}_t$ is the process with lags of $\tilde{y}_t$ replacing lags of y_t, and $\tilde{S}_q$ originates from calculating the autocovariances of $\tilde{y}_t$, a stationary VAR(1) process by Assumption 1. Hence, by the WLLN in Lemma 2,

$$T^{-1}\sum_{t=1}^{T} z_t z_t' \xrightarrow{p} \tilde{S}_q. \tag{39}$$

Because $T^{-1}\sum_{t=1}^{T} z_t z_t'$ is uniformly bounded as shown above, using (38) into (35), we obtain the desired result:

$$\mathbb{E}(\hat{\beta}_h^{LP}) = \beta_h + o(T^{-1/2}). \qquad \square$$

Proof of Theorem 2. We first derive the asymptotic distribution for the LP estimator. Since we already have the LP estimator decomposition in (35), and we know $T^{-1}\sum_{t=1}^{T} z_t z_t' \xrightarrow{p} \tilde{S}_q$ by (39), applying the CLT in Lemma 3 will deliver the asymptotic distribution of $T^{1/2}(\hat{\beta}_h^{LP} - \beta_h)$. Therefore, we now verify the conditions in Lemma 3 for

$$z_t u_{t+h}^h = \sum_{n=0}^{h} \Phi_n' u_{t+h-n} = \sum_{n=0}^{h} z_t\Phi_n'\Gamma(\epsilon_{t+h-n} + T^{-1/2}\alpha(L)\epsilon_{t+h-n}).$$

By Lemma 1, u_t is $\mathcal{L}_d$ NED of size $m^{-1/2}$ on ϵ_t, hence so is u_{t+h}^h by Theorem 17.8 in Davidson (1994). Also, since y_t shares this NED property, so does z_t, and, by Theorem 17.10 in Davidson (1994), so does $z_t u_{t+h}^h$ except it is $\mathcal{L}_2$ NED (actually, $\mathcal{L}_r$, but $r =$

$2 + \delta > 2$, so $\mathcal{L}_2$ suffices). By arguments similar to bounding $\mathbb{E}\|y_t\|^2$ in the proof of Theorem 1, one can show that $\sup_t \mathbb{E}|z_{i,t}u^h_{t+h}|^r < M$, where $r = d/2 = 2 + \delta$. We now derive the long-run variance of $z_t u^h_{t+h}$. To that end, consider first the long-run variance of $y_{j,t}u^h_{t+h}$.

$$\lim_{T\to\infty} \text{Var}(T^{-1/2}\sum_{t=1}^{T} y_{j,t}u^h_{t+h}) = T^{-1}\sum_{t,s=1}^{T} \mathbb{E}(y_{j,t}u^h_{t+h}u^h_{s+h}y_{j,s}).$$

Recall that $y_{j,t} = \tilde{y}_{j,t} + T^{-1/2}\bar{u}_t$, and let $u_t = \tilde{u}_t + \bar{v}_t$, where $\tilde{u}_t = \Gamma\epsilon_t$, and $\bar{v}_t = \Gamma\sum_{l=1}^{\infty}\alpha_\ell\epsilon_{t-\ell}$. By the same arguments as for the proof of Lemma 1, $\tilde{u}_t$, $\tilde{y}_t$, $\bar{u}_t$, $\bar{v}_t$ are $\mathcal{L}_d$ NED of the same size on an α-mixing process (in fact, $\tilde{u}_t$ is also of smaller order because it is a martingale difference by Assumption 1). Therefore, so are their cross-products, but by Theorem 17.10 in Davidson (1994), the relevant NED norm for the cross-products is $\mathcal{L}_{1+\delta/2}$. So, the terms in $\sum_{t,s=1}^{T}\mathbb{E}(y_{j,t}u^h_{t+h}u^h_{s+h}y_{j,s}) = \sum_{n,n^*=0}^{h}\Phi_n'\mathbb{E}(y_{j,t}u_{t+h-n}u^h_{s+h-n^*}y_{j,s})\Phi_{n^*}$, pre-multiplied by $T^{-1/2}$, are necessarily of a smaller order than
$\sum_{t,s=1}^{T}\sum_{n,n^*=0}^{h}\Phi_n'\mathbb{E}(\tilde{y}_{j,t}\tilde{u}_{t+h-n}\tilde{u}^h_{s+h-n^*}y_{j,s})\Phi_{n^*}$.

It remains to show that $\sum_{t,s=1}^{T}\sum_{n,n^*=0}^{h}\Phi_n'\mathbb{E}(\tilde{y}_{j,t}\tilde{u}_{t+h-n}\tilde{u}^h_{s+h-n^*}\tilde{y}_{j,s})\Phi_{n^*}$ exists. By Lemma 1 and Theorem 17.9, the product $y_{j,t}\tilde{u}_{t+h-n}$ is also an $\mathcal{L}_2$ NED sequence on an α-mixing process of size $m^{-r/(r-2)}$, with moments $r+\delta > r$ existing by Assumption 1 and Hölder's inequality. Therefore, setting $r = 2+\delta$ in Theorem 17.7 in Davidson (1994), we obtain that $\sum_{t,s=1}^{T}\sum_{n,n^*=0}^{h}\Phi_n'\mathbb{E}(\tilde{y}_{j,t}\tilde{u}_{t+h-n}\tilde{u}^h_{s+h-n^*}\tilde{y}_{j,s})\Phi_{n^*}$. Hence,

$$\lim_{T\to\infty} \text{Var}(T^{-1/2}\sum_{t=1}^{T} y_{j,t}u^h_{t+h}) = \sum_{n,n^*=0}^{h}\Phi_n'\left[T^{-1}\sum_{t,s=1}^{T} E(\tilde{y}_{j,t}\tilde{u}_{t+h-n}\tilde{u}^h_{s+h-n^*}\tilde{y}_{j,s})\right]\Phi_{n^*} \equiv V_{j,h} < \infty,$$

Therefore, by Lemma 3 (CLT), $T^{-1/2}\sum_{t=1}^{T} y_{j,t}u^h_{t+h} \xrightarrow{d} \mathcal{N}(0, V_{j,h})$.

By similar arguments, $T^{-1/2}\sum_{t=1}^{T} z_t u^h_{t+h} \xrightarrow{d} \mathcal{N}(0, V^{LP}_h)$, where

$$V^{LP}_h = \lim_{T\to\infty} Var\left(T^{-1/2}\sum_{t=1}^{T}\tilde{z}_t\tilde{u}^h_{t+h}\right),$$

where $\tilde{u}^h_{t+h} = \sum_{n=0}^{h} \Phi'_n \tilde{u}_{t+h-n}$. Hence,

$$T^{1/2}(\hat{\beta}^{LP}_h - \beta_h) \xrightarrow{d} \mathcal{N}(0, \tilde{S}_q^{-1} V_h^{LP} \tilde{S}_q^{-1}), \tag{40}$$

where $\Sigma_h^{LP} \equiv \tilde{S}_q^{-1} V_h^{LP} \tilde{S}_q^{-1}$. Equation (40) shows that the asymptotic variance of the LP estimator is unaffected by the misspecification. The VAR estimator has asymptotic bias, as shown in Theorem 1; however, for the limiting variance calculations, the asymptotic bias is irrelevant. Therefore, by similar arguments as for LP,

$$T^{1/2}(\hat{\beta}^{VAR}_h - \beta_h) \xrightarrow{d} \mathcal{N}(aBias_h, \Sigma_h^{VAR}), \tag{41}$$

and Σ_h^{VAR} is the asymptotic variance of standard structural VARs obtained from the data generating process $y_t = Ay_{t-1} + \Gamma\epsilon_t$, thus also unaffected by misspecification. $\square$.

Proof of Theorem 3.

Risk decomposition. For TLP at horizon h, the risk function is

$$\begin{aligned} R_h(\lambda_h) &= T\,\mathbb{E}\Big[\big(\hat{\beta}^{TLP}_h(\lambda_h) - \beta_h\big)^2\Big] \\ &= \underbrace{T\big(\mathbb{E}[\hat{\beta}^{TLP}_h(\lambda_h)] - \beta_h\big)^2}_{\text{bias}^2(\lambda_h)} + \underbrace{T\,\mathrm{Var}[\hat{\beta}^{TLP}_h(\lambda_h) - \beta_h]}_{\text{variance}(\lambda_h)}. \end{aligned} \tag{42}$$

Let $b_h = \mathrm{plim}[v(\lambda_h)]$, which exists for each λ_h since $T^{-1}X'X \xrightarrow{p} E(x_t x_t')$ by Assumptions 1-2 and the LLN for α-mixing processes. Moreover, by the LLN, $\sup_{\lambda_h \in [0,\infty)} |v(\lambda_h) - b_h| \xrightarrow{p} 0$. From Theorem 1, the squared bias is:

$$\begin{aligned} T\big(\mathbb{E}[\hat{\beta}^{TLP}_h(\lambda_h)] - \beta_h\big)^2 &= T\big(\mathbb{E}[(v(\lambda_h)\hat{\beta}^{LP}_h + (1 - v(\lambda_h))\hat{\beta}^{VAR}_h - \beta_h]\big)^2 \\ &= T\big(\mathbb{E}[v(\lambda_h)(\hat{\beta}^{LP}_h - \beta_h)] + \mathbb{E}[(1 - v(\lambda_h))(\hat{\beta}^{VAR}_h - \beta_h)]\big)^2 \\ &= b_h^2\,(\mathbb{E}(T^{1/2}[\hat{\beta}^{LP}_h - \beta_h]))^2 + (1 - b_h)^2\,(\mathbb{E}(T^{1/2}[\hat{\beta}^{VAR}_h - \beta_h]))^2 \\ &+ 2b_h\,(1 - b_h)\,\mathbb{E}[T^{1/2}(\hat{\beta}^{LP}_h - \beta_h)]\,\mathbb{E}[T^{1/2}(\hat{\beta}^{VAR}_h - \beta_h)] + o(1) \\ &= b_h^2\, \times 0 + (1 - b_h)^2\, C_h^2 + 2b_h\,(1 - b_h) \times 0 \times C_h + o(1) \\ &= (1 - b_h)^2 C_h^2 + o(1). \end{aligned} \tag{43}$$

where $C_h = \text{aBias}_h$. From Theorem 2, the variance term in the risk criterion is:

$$
\begin{aligned}
T\text{Var}[\hat{\beta}_h^{TLP}(\lambda_h)] &= T\text{Var}[b_h\hat{\beta}_h^{LP}(\lambda_h) + (1-b_h)\hat{\beta}_h^{VAR}(\lambda_h)] \\
&= b_h^2\,\text{Var}(T^{1/2}[\hat{\beta}_h^{LP}(\lambda_h) - \beta_h]) + (1-b_h)^2\,\text{Var}(T^{1/2}[\hat{\beta}_h^{VAR}(\lambda_h) - \beta_h]) \\
&\quad + 2\,b_h\,(1-b_h)\,\text{Cov}(T^{1/2}[\hat{\beta}_h^{LP}(\lambda_h) - \beta_h], T^{1/2}[\hat{\beta}_h^{VAR}(\lambda_h) - \beta_h]\,) + o(1) \\
&= b_h^2\,\Sigma_h^{LP} + (1-b_h)^2\,\Sigma_h^{VAR} + 2\,b_h\,(1-b_h)\,\Sigma_h^{COV}. \qquad (44)
\end{aligned}
$$

Substituting (43)-(44) into (42), we obtain:

$$
R_h(\lambda_h) = (1-b_h)^2\,C_h^2 + \,[b_h^2\,\Sigma_h^{LP} + (1-b_h)^2\,\Sigma_h^{VAR} + \,2b_h\,(1-b_h)\,\Sigma_h^{COV}] + o(1). \qquad (45)
$$

Empirical risk criterion. From Theorem 2,

$$
\hat{C}_h = T^{1/2}\big(\hat{\beta}_h^{LP} - \hat{\beta}_h^{VAR}\big) \xrightarrow{d} \mathcal{N}(-C_h, \Sigma_h^{LP} + \Sigma_h^{VAR} - 2\Sigma_h^{COV}).
$$

It follows that

$$
\mathbb{E}[\hat{C}_h^2] = C_h^2 + \Sigma_h^{LP} + \Sigma_h^{VAR} - 2\Sigma_h^{COV} + o(1). \qquad (46)
$$

Hence, $C_h^2 = \mathbb{E}[\hat{C}_h^2] - \Sigma_h^{LP} - \Sigma_h^{VAR} + 2\Sigma_h^{COV} + o(1)$. Since $\hat{\Sigma}_h^{\mu} \xrightarrow{p} \Sigma_h^{\mu}$ for $\mu \in \{LP, VAR, COV\}$, combining the estimated bias and variance terms, the asymptotically unbiased empirical risk criterion is:

$$
\begin{aligned}
\hat{R}_h(\lambda_h) &= T(1-v(\lambda_h))^2(\hat{\beta}_h^{LP} - \hat{\beta}_h^{VAR})^2 - (1-v(\lambda_h))^2\hat{\Sigma}_h^{LP} - (1-v(\lambda_h))^2\hat{\Sigma}_h^{VAR} \\
&\quad + 2(1-v(\lambda_h))^2\hat{\Sigma}_h^{COV} + v(\lambda_h)^2\hat{\Sigma}_h^{LP} + (1-v(\lambda_h))^2\hat{\Sigma}_h^{VAR} \\
&\quad + 2v(\lambda_h)(1-v(\lambda_h))\hat{\Sigma}_h^{COV} \\
&= T(1-v(\lambda_h))^2(\hat{\beta}_h^{LP} - \hat{\beta}_h^{VAR})^2 + (2v(\lambda_h)-1)\hat{\Sigma}_h^{LP} + 2(1-v(\lambda_h))\hat{\Sigma}_h^{COV}. \qquad (47)
\end{aligned}
$$

Since $\hat{R}_h(\lambda_h)$ is continuous in λ_h, $\sup_{\lambda_h \in (0,\infty)} |\mathbb{E}[\hat{R}_h(\lambda_h)] - R_h(\lambda_h)| \to 0$. $\square$

Proof of Theorem 4. Since there is a one-to-one mapping between λ_h and $v(\lambda_h) = (X'X + \lambda_h)^{-1}X'X$, optimizing $\hat{R}_h(\lambda_h)$ with respect to λ_h is equivalent to optimizing it with respect to $v(\lambda_h)$. Let $\hat{A}_h = T(\hat{\beta}_h^{LP} - \hat{\beta}_h^{VAR})^2$. Then

$$\begin{aligned}
\frac{\partial \hat{R}_h(\lambda_h)}{\partial v(\lambda_h)} &= -2(1 - v(\lambda_h))\hat{A}_h + 2(\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{COV}) = 0 \\
\Leftrightarrow v(\lambda_h)\hat{A}_h &= \hat{A}_h - (\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{COV}) \\
\Leftrightarrow v(\lambda_h) &= 1 - \frac{(\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{COV})}{\hat{A}_h}.
\end{aligned}$$

Since $\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{COV} > 0$, the weights are in finite samples less than one. However, they also need to be positive according to the definition $v(\lambda_h) = (X'X + \lambda_h)^{-1}X'X$. Therefore, the optimal weights are:

$$v(\hat{\lambda}_h) = \max\left[1 - \frac{\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{COV}}{\hat{A}_h}, 0\right].$$

with $\hat{\lambda}_h \to \infty$ when $v(\hat{\lambda}_h) = 0$, and

$$T^{-1}\hat{\lambda}_h = \frac{T^{-1}X'X(\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{COV})}{\hat{A}_h - (\hat{\Sigma}_h^{LP} - \hat{\Sigma}_h^{COV})}.$$

otherwise. □

C Additional simulations

C.1 Larger misspecification η

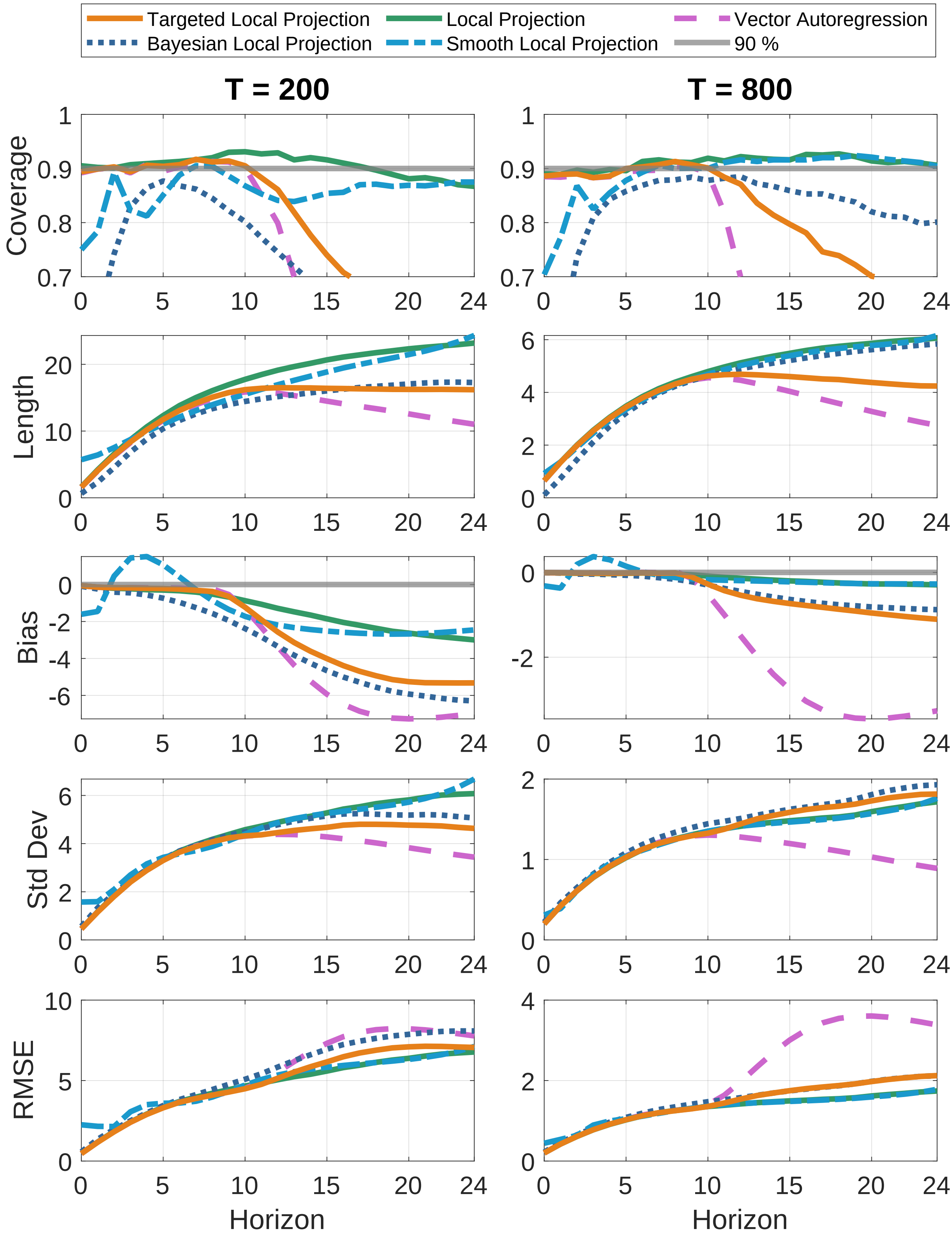

Figure C1: Coverage, length, bias, standard deviation, and RMSE for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 8$, conditionally homoskedastic errors. Methods: Targeted Local Projections (10,8), Local Projections (10), Vector Autoregression (8), Smooth Local Projections (10) and Bayesian Local Projections (8,8). Inference by MSDB for TLP, LP, VAR and SLP.

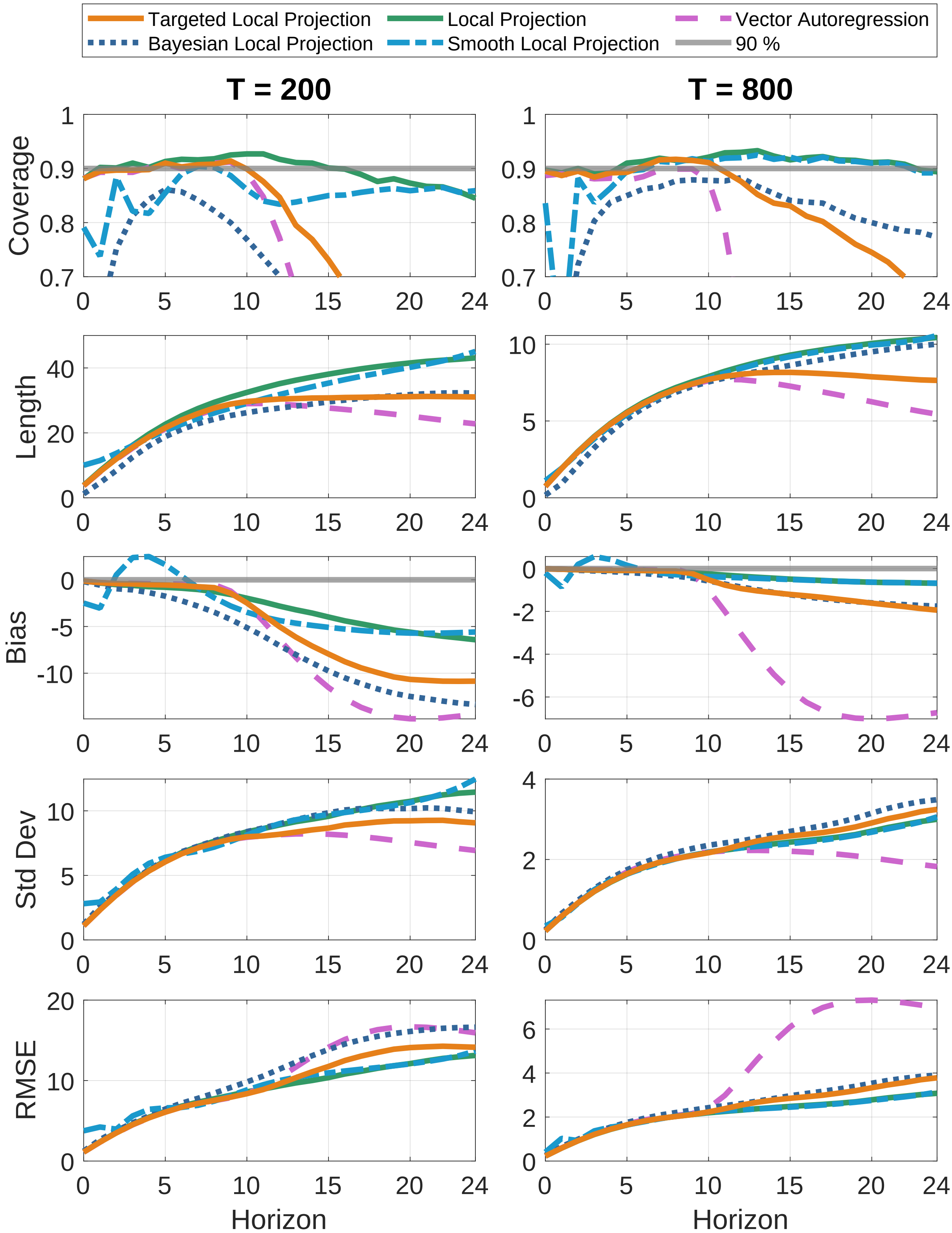

Figure C2: Coverage, length, bias, standard deviation, and RMSE for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 16$, conditionally homoskedastic errors. Methods: Targeted Local Projections (10,8), Local Projections (10), Vector Autoregression (8), Smooth Local Projections (10) and Bayesian Local Projections (8,8). Inference by MSDB for TLP, LP, VAR and SLP.

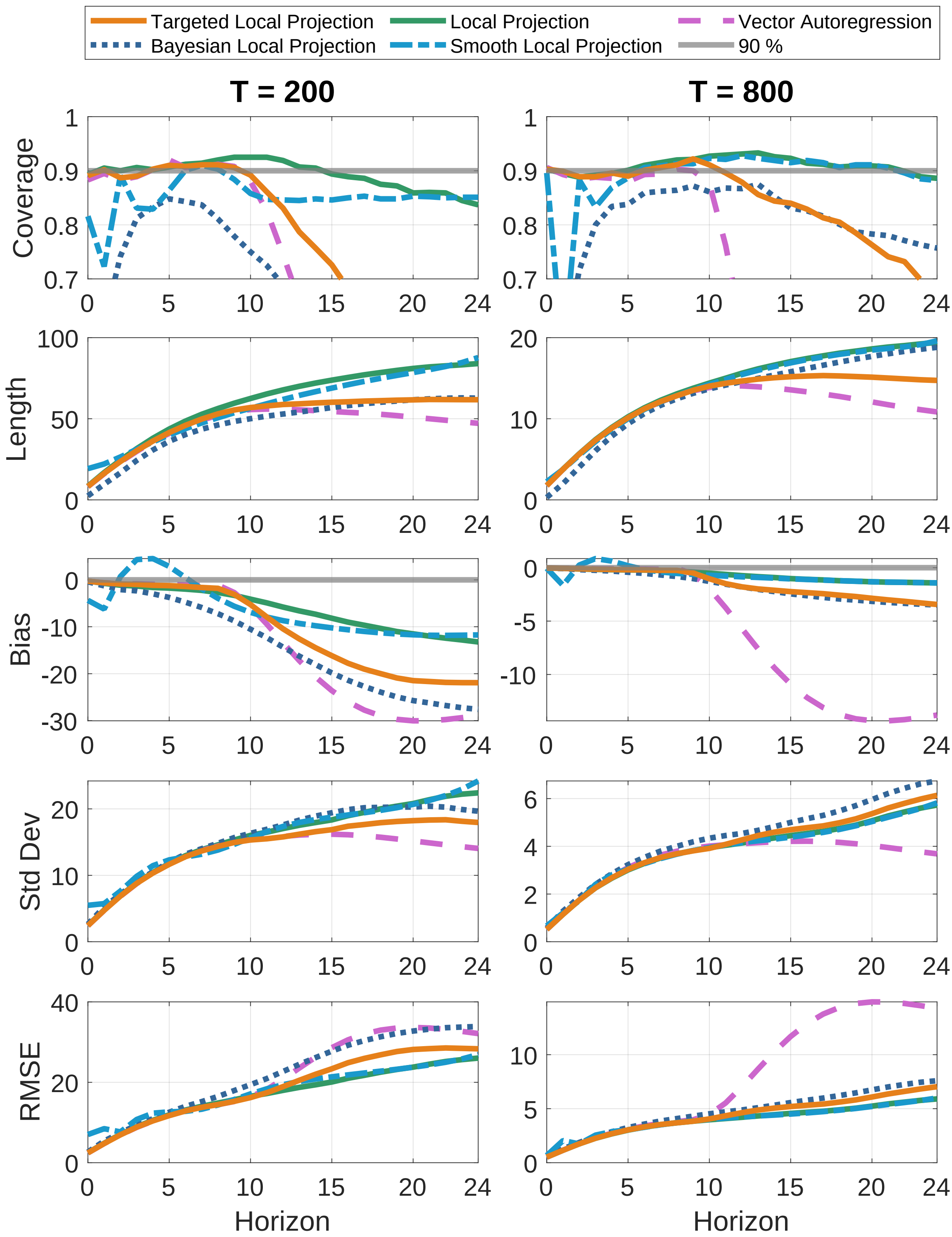

Figure C3: Coverage, length, bias, standard deviation, and RMSE for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 32$, conditionally homoskedastic errors. Methods: Targeted Local Projections (10,8), Local Projections (10), Vector Autoregression (8), Smooth Local Projections (10) and Bayesian Local Projections (8,8). Inference by MSDB for TLP, LP, VAR and SLP.

C.2 GARCH(1,1) errors

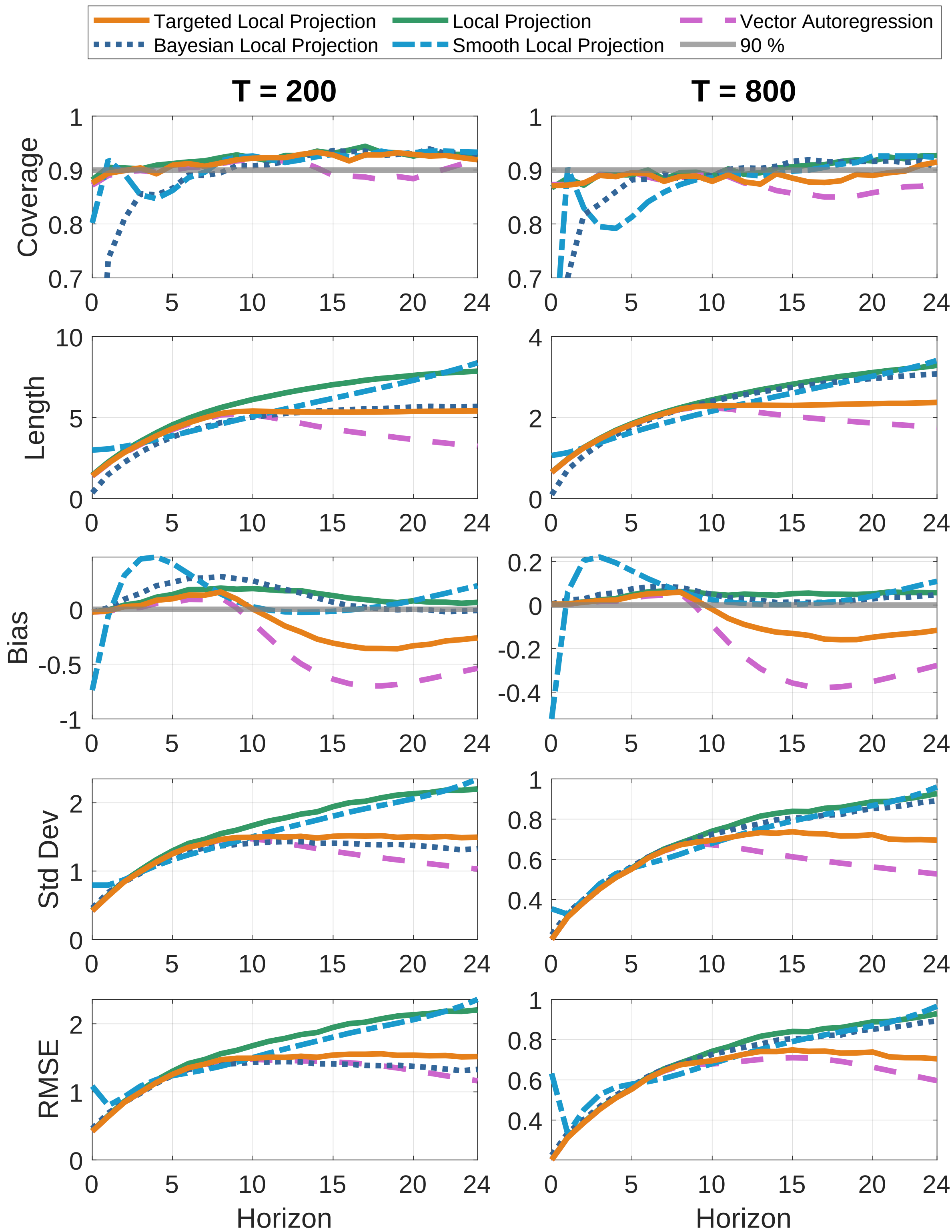

Figure C4: Coverage, length, bias, standard deviation, and RMSE for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 1$, **GARCH(1,1) errors**. Methods: Targeted Local Projections (10,8), Local Projections (10), Vector Autoregression (8), Smooth Local Projections (10) and Bayesian Local Projections (8,8). Inference by MSDB for TLP, LP, VAR and SLP.

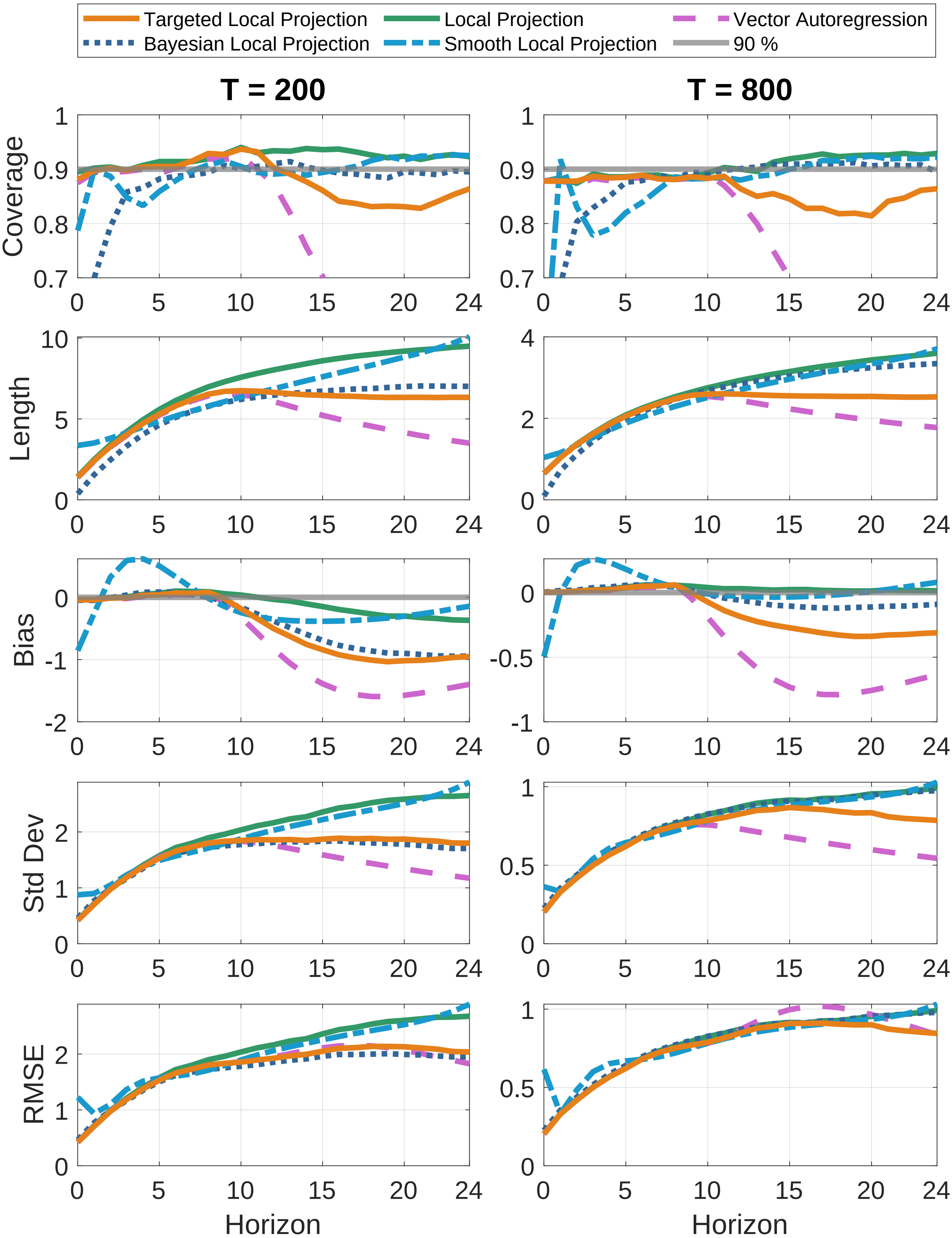

Figure C5: Coverage, length, bias, standard deviation, and RMSE for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 2$, **GARCH(1,1) errors**. Methods: Targeted Local Projections (10,8), Local Projections (10), Vector Autoregression (8), Smooth Local Projections (10) and Bayesian Local Projections (8,8). Inference by MSDB for TLP, LP, VAR and SLP.

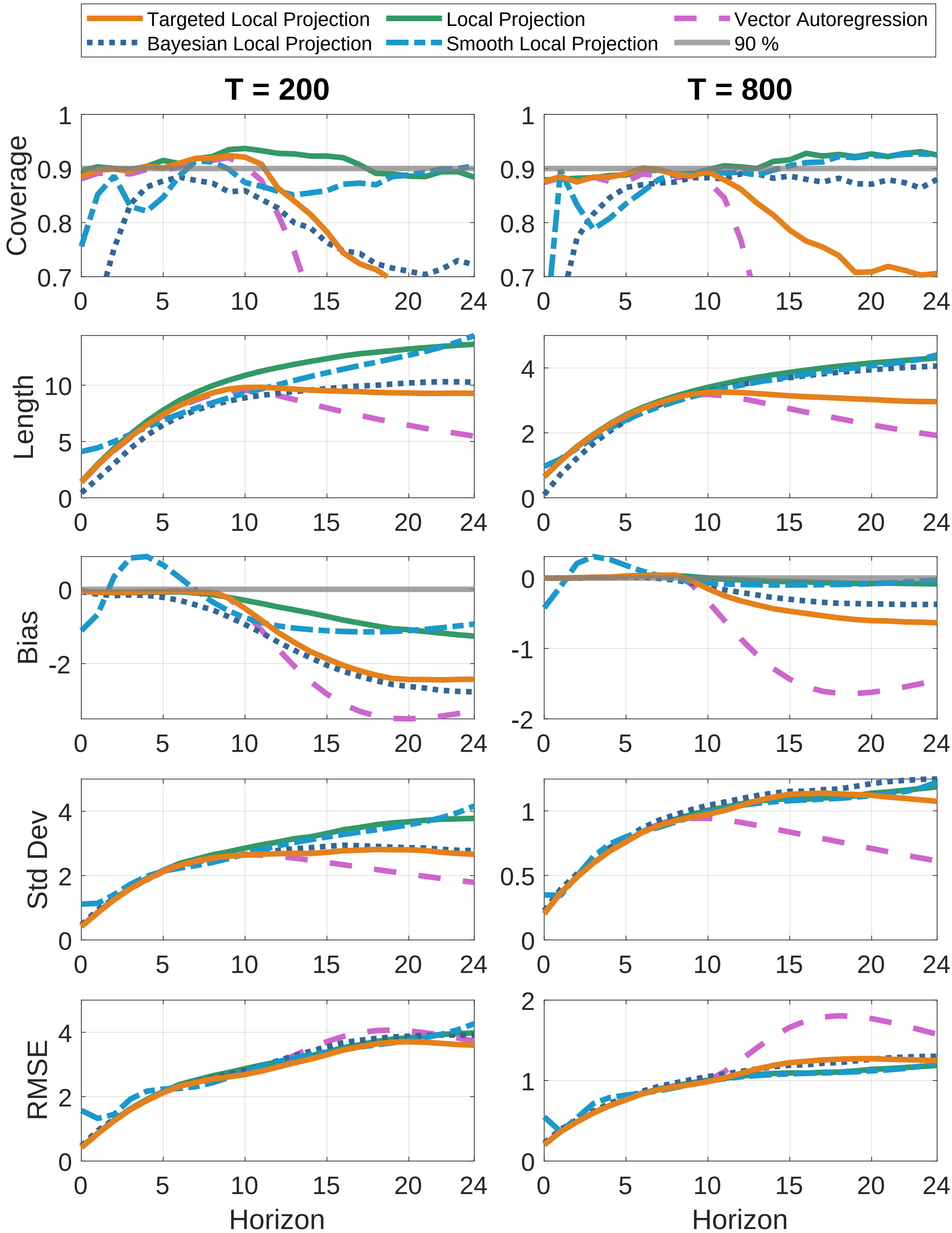

Figure C6: Coverage, length, bias, standard deviation, and RMSE for $T = 200$ (left) and $T = 800$ (right), DGP = VARMA(1,100), $\eta = 4$, **GARCH(1,1) errors**. Methods: Targeted Local Projections (10,8), Local Projections (10), Vector Autoregression (8), Smooth Local Projections (10) and Bayesian Local Projections (8,8). Inference by MSDB for TLP, LP, VAR and SLP.

D Additional empirical results

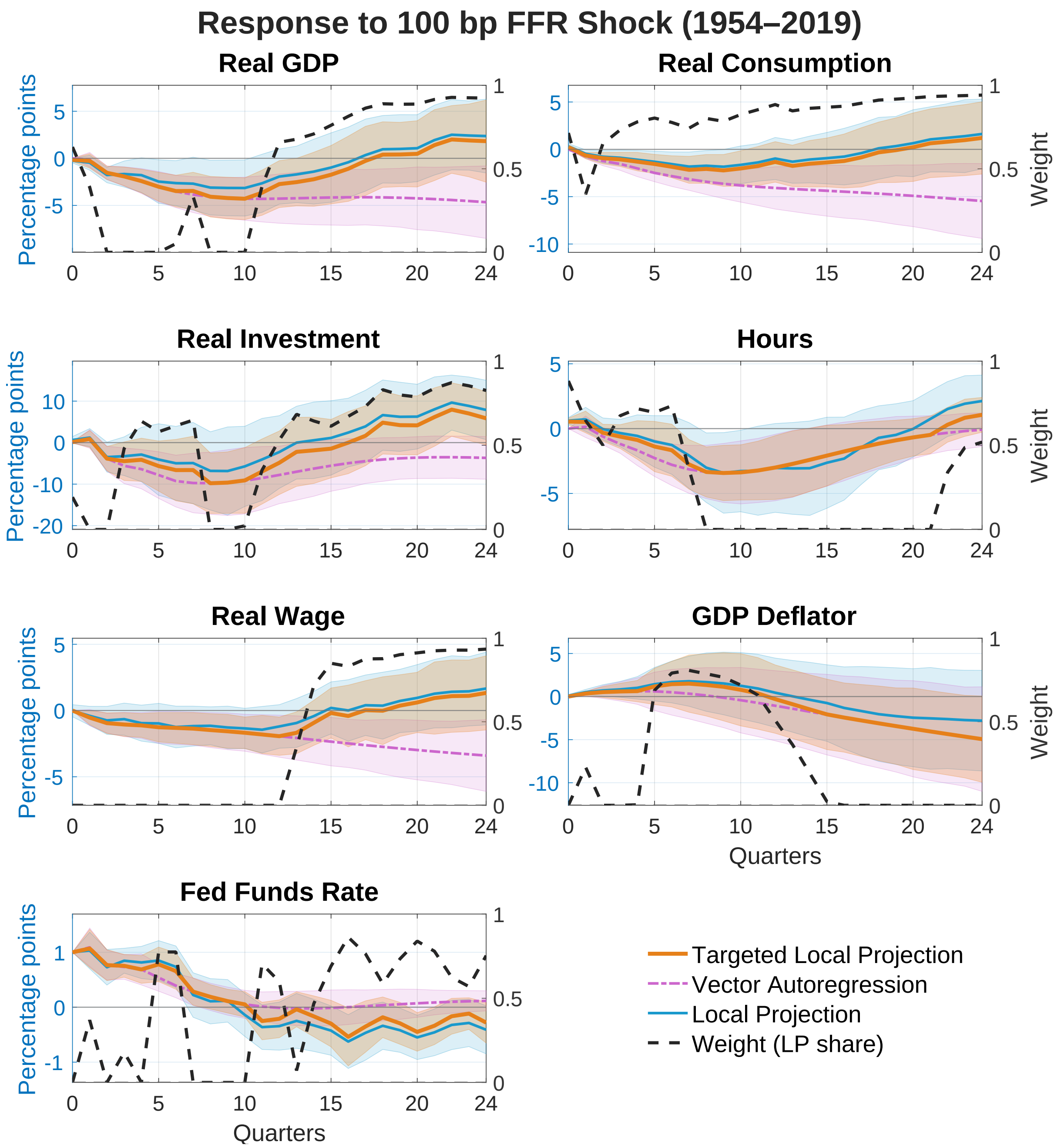

Figure D1: Impulse responses to a 100 basis point shock to the federal funds rate (1954–2019). The panels display the responses of Real GDP, Real Consumption, Real Investment, Hours, Real Wage, GDP Deflator, and the federal funds rate. Shaded regions correspond to 90% confidence intervals. Estimation uses $p = 8$ and $q = 4$ lags.